A2 · Unit 4 · WBI14 Energy, Environment, Microbiology and Immunity

Photosynthesis

Photosynthesis is how green plants, algae and some bacteria turn light energy into chemical energy stored in sugars. Almost every food chain starts here, and the oxygen in the air is a by-product of the process. This page follows the energy from the moment light strikes a pigment molecule to the sugars made in the stroma, and explains why the rate of the process rises and falls.

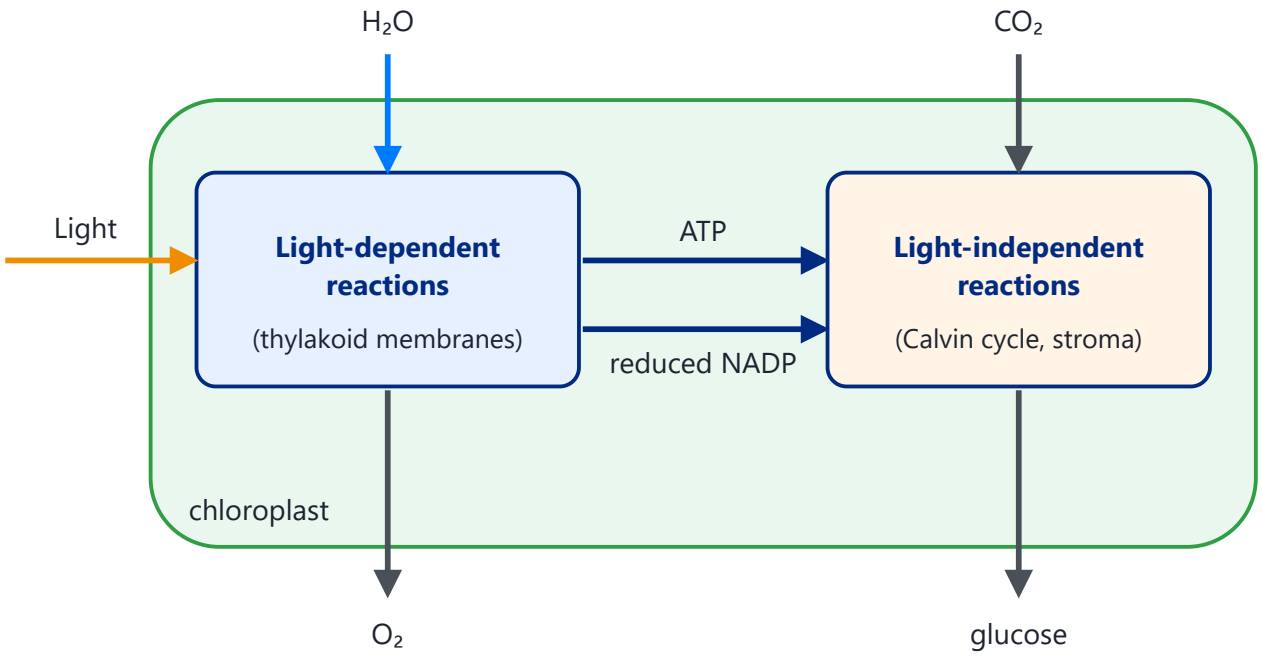
1. The overall process and its two stages

The bonds holding hydrogen to oxygen in water are strong, so splitting water needs a large input of energy. In photosynthesis that energy comes from light. The hydrogen removed from water is not released as a gas. It is combined with carbon dioxide to make glucose, which acts as a fuel because the energy used to make it can be released again in respiration. The oxygen left over from the water is released into the atmosphere.

The balanced equation is $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6H_2O + 6O_2$, with light energy and chlorophyll as the conditions. Water appears on both sides because all the oxygen released comes from water, not from carbon dioxide, so twelve water molecules must be split to release six O_2. Cancelling the water gives the simpler form $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$.

The process is a chain of energy changes. Light energy is converted into the energy of excited electrons, and then into chemical energy held in ATP and reduced NADP. That chemical energy is finally used to make the bonds of glucose. Because these steps need different conditions and different places in the chloroplast, photosynthesis is divided into two linked stages.

- **Light-dependent stage:** takes place on the thylakoid membranes. It needs light, splits water (photolysis), releases oxygen and makes ATP and reduced NADP.
- **Light-independent stage (Calvin cycle):** takes place in the stroma. It uses the ATP and reduced NADP to reduce carbon dioxide to sugar. It does not use light directly, but it stops soon after the products of the light-dependent stage run out.

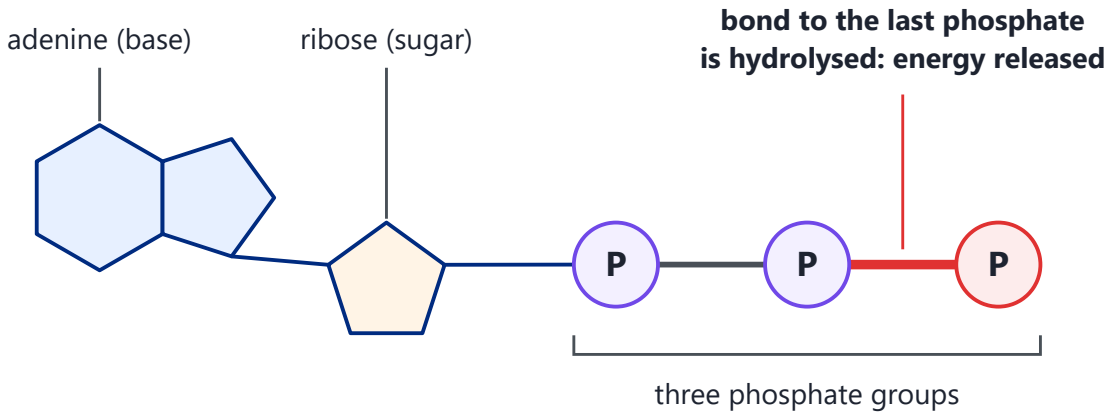


The two stages are linked: ATP and reduced NADP flow from the light-dependent reactions to the light-independent reactions.

2. ATP: the energy carrier

ATP (adenosine triphosphate) is a nucleotide made of the base adenine, the 5-carbon sugar ribose and a chain of three phosphate groups. Cells use it as their immediate source of energy. Removing the last phosphate group by hydrolysis forms ADP and inorganic phosphate (Pi) and releases energy that can drive other reactions, such as muscle contraction, active transport and the synthesis of large molecules. The reaction is catalysed by the enzyme ATPase.

The reverse reaction, joining ADP and Pi to remake ATP, is phosphorylation. It needs an input of energy and is catalysed by ATP synthase. In photosynthesis that energy comes from light, so the process is called photophosphorylation. In respiration it comes from the oxidation of organic molecules.



ATP is adenine and ribose attached to three phosphate groups; hydrolysis of the last phosphate releases energy.

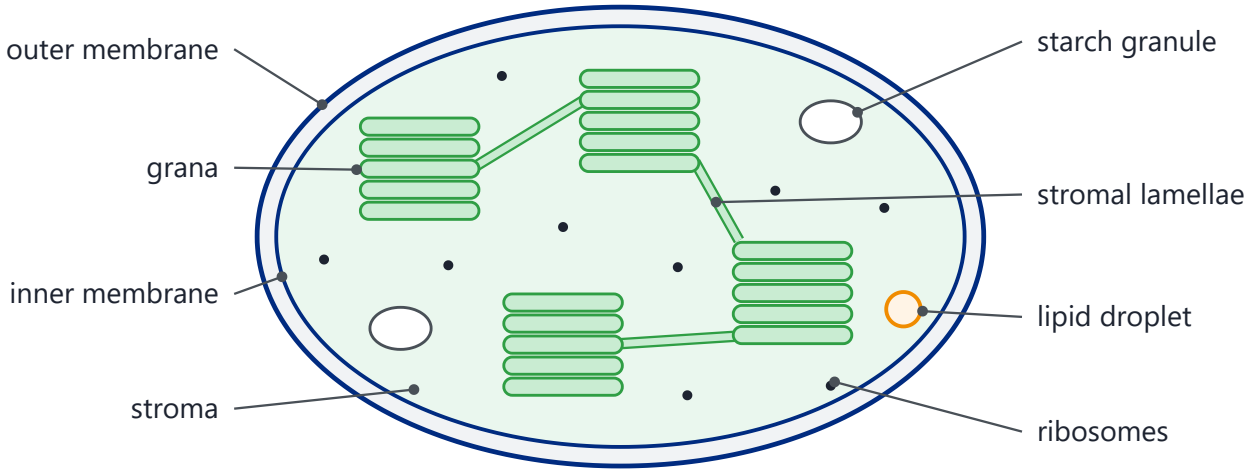
- **Hydrolysis of ATP:** ATP + H_2O → ADP + Pi, releasing energy for biological processes.
- **Phosphorylation of ADP:** ADP + Pi → ATP, which requires energy.
- **Why ATP suits its job:** one hydrolysis releases a small, usable amount of energy in a single step, the reaction is fast, and ATP is soluble, so it moves easily to where energy is needed and is quickly remade.

3. Chloroplast structure and function

The chloroplast is an organelle found in the cells of green plant tissues, especially the palisade mesophyll. Each part of its structure suits one of the two stages. It is bounded by an envelope of two membranes. The outer membrane is freely permeable to small molecules such as carbon dioxide and water. The inner membrane holds transport proteins that control what enters and leaves, including sugars and proteins made in the cytoplasm but used inside the chloroplast.

The fluid inside the envelope is the stroma. It contains the enzymes of the Calvin cycle, including RUBISCO, so it is the site of the light-independent reactions. The stroma also contains starch grains that store the products, lipid droplets, a loop of DNA and 70S ribosomes, so the chloroplast can make some of its own proteins.

Running through the stroma is a system of flattened membrane sacs called thylakoids, the site of the light-dependent reactions. Thylakoids are stacked into grana (singular granum), and the grana are linked by flat membranes called lamellae. The thylakoid membranes hold the photosystems, the electron carriers and ATP synthase. Stacking gives a very large membrane area for capturing light. Each thylakoid also encloses a small space, the lumen, where hydrogen ions can build up to a high concentration, which is essential for making ATP.



The stroma is the site of the light-independent reactions, and the thylakoid stacks (grana) are the site of the light-dependent reactions.

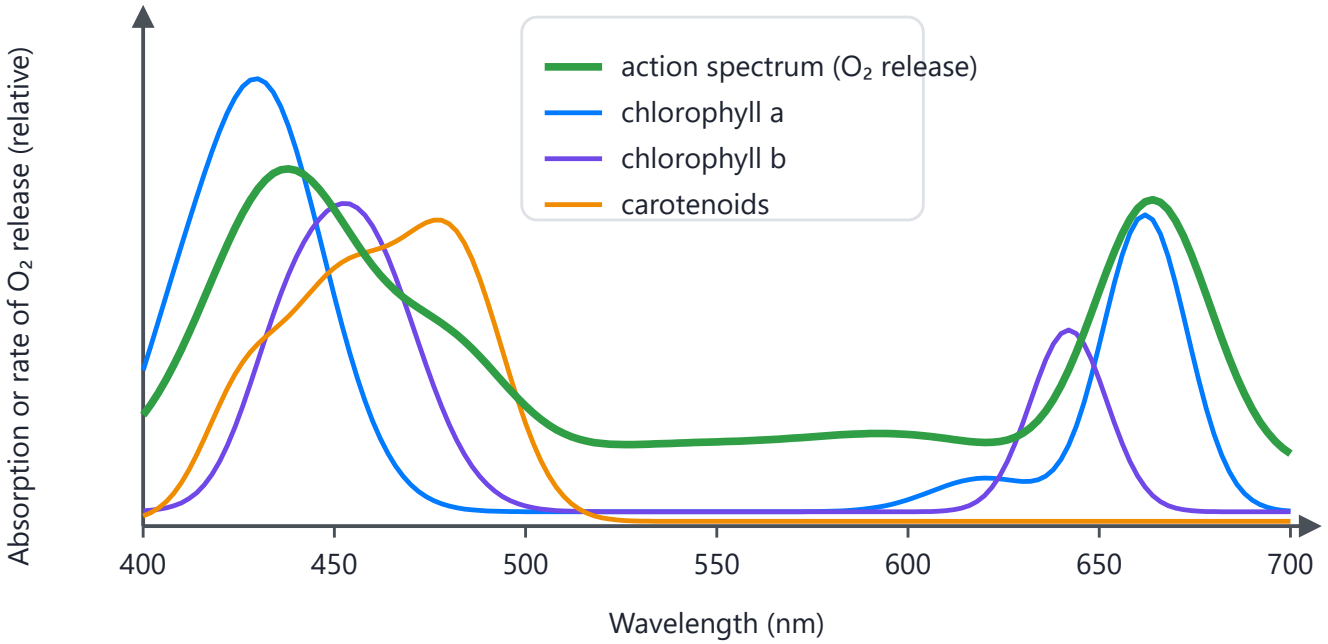
EXAM TIP

When asked how a chloroplast is adapted, link each structure to a stage: thylakoid membranes with photosystems, electron carriers and ATP synthase for the light-dependent reactions; a lumen in which H^+ accumulates; and a stroma containing RUBISCO and other Calvin cycle enzymes.

4. Pigments, spectra and chromatography

Chlorophyll a is the main pigment. It absorbs most strongly in the blue-violet and red parts of the spectrum and reflects green light, which is why leaves look green. Chlorophyll b and the carotenoids (carotene and xanthophylls) are accessory pigments. They absorb wavelengths that chlorophyll a absorbs poorly and pass the energy to chlorophyll a, so together the pigments use more of the visible spectrum. Each chlorophyll molecule has a head that absorbs light and a long hydrocarbon tail that anchors it in the thylakoid membrane.

Two graphs describe this. An absorption spectrum shows how much light of each wavelength a pigment absorbs. An action spectrum shows how effective each wavelength is at driving photosynthesis, measured for example as the rate of oxygen release. The action spectrum closely follows the combined absorption spectra of the pigments, which is evidence that the light absorbed by these pigments is the light used in photosynthesis.



The action spectrum (green) has peaks in the blue and red, matching the absorption peaks of the pigments.

The pigments in a leaf extract can be separated by chromatography on paper or on a thin-layer plate. A concentrated extract is spotted onto a pencil line near the bottom, and the bottom edge is dipped in a solvent that is kept below the spot. As the solvent rises, each pigment moves at its own speed, depending on how soluble it is in the solvent and how strongly it is held by the paper or plate. In the usual solvents carotene travels furthest, followed by xanthophyll, chlorophyll a and then chlorophyll b.

CORE PRACTICAL

Core practical: separating chloroplast pigments. Grind leaves with a little propanone to extract the pigments. Build up a small, dense spot by adding drops and letting each one dry. Use a pencil line, because ink would itself separate. Remove the paper before the solvent reaches the top and mark the solvent front at once. Mark the pigment spots quickly too, as some fade in light.

Calculating an Rf value

1. Measure from the pencil line to the solvent front. Suppose this is 10.0 cm.
2. Measure from the pencil line to the centre of the pigment spot. Suppose this is 6.5 cm.

3 Rf = distance moved by the pigment ÷ distance moved by the solvent = 6.5 ÷ 10.0 = 0.65.

An Rf value is a ratio, so it does not depend on how far the solvent ran. It always lies between 0 and 1, and for a given pigment, solvent and paper it is constant, so it can be compared with known values to identify the pigment.

You can run a virtual chromatogram below and measure the Rf value of each spot.

5. Photosystems

On the thylakoid membrane, the pigments are grouped with proteins into photosystems. Each photosystem has an antenna complex of many pigment molecules. Light energy absorbed by any of them is passed from molecule to molecule until it reaches the reaction centre, a pair of chlorophyll a molecules. There the energy excites an electron so much that it leaves the chlorophyll and is taken up by an electron acceptor. The antenna works like a funnel, collecting light of many wavelengths and concentrating the energy on one place.

There are two photosystems. The reaction centre of photosystem II absorbs best at 680 nm, so it is called P680. The reaction centre of photosystem I absorbs best at 700 nm and is called P700. The numbers record the order of discovery, not the order of use: in the main pathway, photosystem II acts first.

Feature	Photosystem II	Photosystem I
Reaction centre	P680	P700
Main position	Inner surfaces of the grana	Lamellae and the outer surfaces of the grana
Replaces its lost electrons from	Photolysis of water	Electrons passed along the carriers from photosystem II
Pathways it takes part in	Non-cyclic only	Both cyclic and non-cyclic

6. The light-dependent reactions

Photolysis: splitting water

When P680 loses an electron, it must gain another before it can absorb light again. The replacement comes from water. Using light energy, an enzyme complex on the inside of the thylakoid membrane splits water: $2H_2O \rightarrow 4H^+ + 4e^- + O_2$. The electrons go to photosystem II, the hydrogen ions stay in the lumen, and the oxygen is a waste product that diffuses out of the leaf or is used in respiration.

Non-cyclic photophosphorylation

Non-cyclic photophosphorylation is the main pathway. It uses both photosystems and produces ATP, reduced NADP and oxygen. The steps below follow the electrons from water to NADP.

Following the electrons through non-cyclic flow

1. Light absorbed by the antenna of photosystem II excites electrons in P680. They leave the chlorophyll and pass to an electron acceptor.

2. Photolysis of water replaces the lost electrons, releasing oxygen and adding hydrogen ions to the lumen.

   Without this supply of electrons, P680 could not absorb light again and the flow would stop.

3. The electrons are passed along a chain of electron carriers in the thylakoid membrane in a series of redox reactions, losing energy at each step.

4. Some of that energy is used to pump hydrogen ions from the stroma across the thylakoid membrane into the lumen.

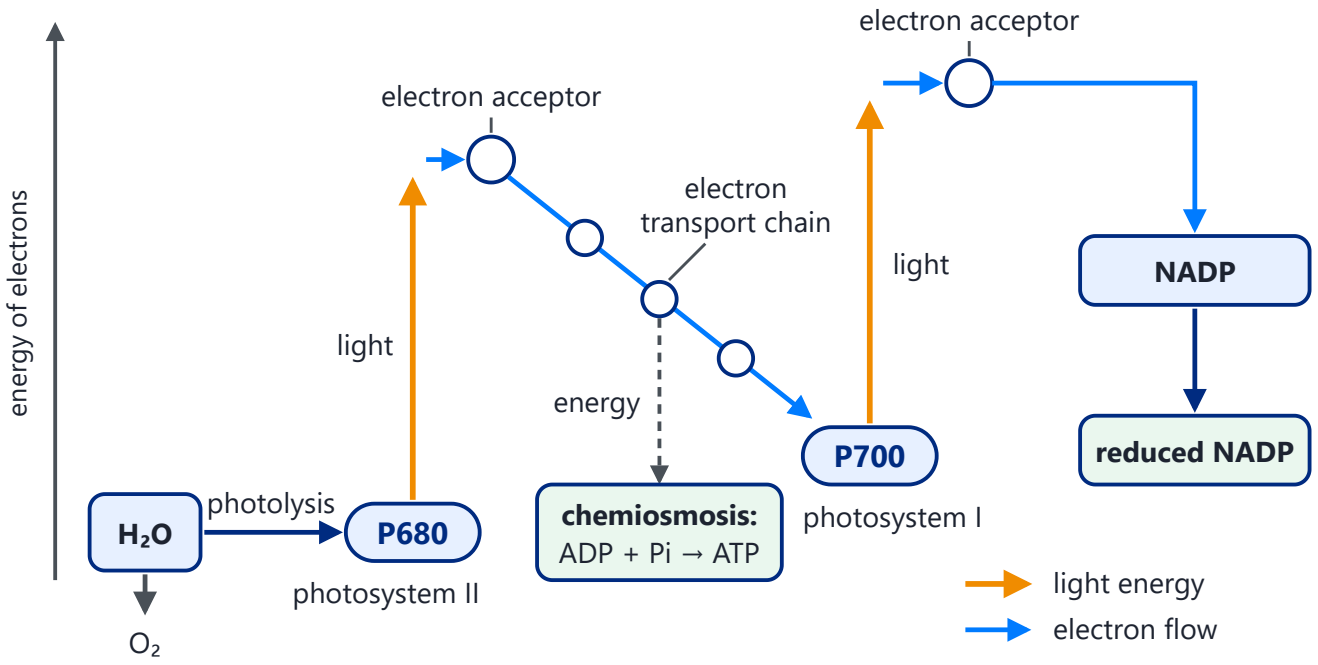
   Moving ions against their concentration gradient needs energy, which comes from the electrons.

5. Hydrogen ions are now much more concentrated in the lumen than in the stroma. They can only diffuse back through channels in ATP synthase.

6. The flow of hydrogen ions through ATP synthase provides the energy to join ADP and Pi into ATP. Making ATP this way is chemiosmosis, and because light drives it, the process is photophosphorylation.

7. The electrons reach photosystem I, where light excites them again from P700 to a higher energy level, and they pass to a second electron acceptor.

8. The enzyme NADP reductase uses these electrons, together with hydrogen ions from the stroma, to reduce NADP to reduced NADP. Reduced NADP carries hydrogen to the Calvin cycle.



Non-cyclic flow: electrons travel from water through photosystem II, the carrier chain and photosystem I to NADP.

COMMON MISTAKE

The light-dependent reactions do not make glucose. Their products are ATP, reduced NADP and oxygen. Glucose is made only in the light-independent reactions.

Cyclic photophosphorylation

Cyclic photophosphorylation uses photosystem I alone. Electrons excited from P700 pass to an acceptor and along the carrier chain, which pumps hydrogen ions into the lumen, so ATP is made by chemiosmosis. Instead of reducing NADP, the electrons return to P700 and the cycle repeats. Because water is not split and NADP is not reduced, the only product is ATP.

Cyclic flow matters because the Calvin cycle uses more ATP than reduced NADP: 3 ATP for every 2 reduced NADP. Non-cyclic flow on its own does not provide enough ATP, so cyclic flow tops up the supply.

7. The light-independent reactions (Calvin cycle)

The Calvin cycle takes place in the stroma. It reduces carbon dioxide to a 3-carbon sugar using the ATP and reduced NADP from the light-dependent reactions, and then rebuilds the molecule that accepts carbon dioxide so the cycle can continue. Every step is catalysed by an enzyme, so the rate depends on temperature as well as on the supply of carbon dioxide, ATP and reduced NADP.

One turn of the Calvin cycle

1. Carbon dioxide diffuses into the leaf through the stomata and into the stroma. The enzyme RUBISCO catalyses its combination with ribulose bisphosphate (RuBP), a 5-carbon compound. This is carbon fixation.
2. The 6-carbon product is unstable and immediately splits into two molecules of glycerate 3-phosphate (GP), a 3-carbon compound.
3. GP is reduced to glyceraldehyde 3-phosphate (GALP). Reduced NADP supplies the hydrogen and ATP supplies the energy, one of each for every GP.

   This is the reduction step. Adding hydrogen stores chemical energy in the product.
4. Of every 12 GALP formed, 10 are used to regenerate 6 RuBP, which uses a further 6 ATP. The other 2 GALP leave the cycle and can form one glucose or other organic compounds.
5. Fixing 6 CO_2 therefore uses 18 ATP (12 for reduction and 6 for regeneration) and 12 reduced NADP. The ADP, Pi and NADP return to the thylakoids to be recharged.

   Without a constant supply of ATP and reduced NADP, RuBP cannot be regenerated and carbon fixation soon stops.

KEY IDEA

For one glucose: 6 CO_2 + 18 ATP + 12 reduced NADP in; 1 glucose, 18 ADP + 18 Pi and 12 NADP out.

EXAM TIP

"Light-independent" does not mean "happens in the dark". The reactions continue only briefly in darkness, while stored ATP and reduced NADP last. Say that the stage depends on the products of the light-dependent stage rather than on light itself.

8. What the products are used for

GALP is the starting point for nearly every organic molecule in a plant, and through food chains for those in animals, fungi and most other organisms too. Two GALP can be combined to make glucose, which is used in respiration to supply ATP. Glucose can be joined with fructose to make sucrose, the form in which sugar is carried in the phloem. Alpha glucose units are linked by glycosidic bonds into starch, a store made of amylose (unbranched, 1,4 bonds) and amylopectin (branched, with 1,4 and 1,6 bonds). Beta glucose units are linked into the long unbranched chains of cellulose in cell walls.

Lipids are made from glycerol, which comes from GALP, and fatty acids, which come from GP. Amino acids are made from these 3-carbon compounds combined with nitrogen from nitrate ions taken up by the roots, and some also need sulfur. Nucleic acids need a pentose sugar made from GALP, nitrogen-containing bases made using nitrate, and phosphate from the soil. This is why mineral ions as well as photosynthesis are needed for growth.

9. Factors that limit the rate

The rate of photosynthesis depends mainly on light intensity, carbon dioxide concentration and temperature. At any moment, the rate is limited by whichever factor is furthest from its optimum, the limiting factor. Only an increase in the limiting factor will raise the rate.

Light intensity

At low light intensity, the rate rises in proportion to the light, because more electrons are excited and more ATP and reduced NADP are made. The curve then levels off: light is no longer limiting, and carbon dioxide or temperature now limits the rate. Very intense light can damage the chloroplasts.

Carbon dioxide concentration

Carbon dioxide is the substrate that RUBISCO fixes, so raising its concentration increases the rate until another factor becomes limiting. In air it is often the main limiting factor, because its concentration is low, about 0.04%.

Temperature

Temperature acts mainly on the enzymes of the Calvin cycle, such as RUBISCO. Below the optimum, a rise in temperature gives molecules more kinetic energy, so enzyme and substrate collide more often and more enzyme-substrate complexes form. Above the optimum, bonds holding the enzyme in shape break, the active site changes shape and the enzyme denatures, so the rate falls. The optimum differs between species and is often around 25 °C in temperate plants.

Identifying the limiting factor on a graph

1. Find the part of the curve that is still rising. In that region the factor on the x-axis is limiting.
2. Find where the curve levels off. Beyond that point the x-axis factor is no longer limiting.
3. Use the other conditions in the question to name the factor that now limits the rate. If a second curve at a higher CO_2 concentration or temperature levels off higher, that factor was limiting the first curve.

Examiners want the factor that is limiting, not just the factor being changed on the axis.

Practise this on the graph below, where you set the CO_2 concentration and temperature yourself.

What happens to GP and RuBP when conditions change

The levels of the Calvin cycle intermediates show which step has been affected.

Change	GP	RuBP	Reason
Light intensity falls	Rises	Falls	Less ATP and reduced NADP are made, so GP cannot be reduced to GALP and builds up. Less GALP means RuBP cannot be regenerated, but it is still used to fix CO_2, so its level falls.
Carbon dioxide concentration falls	Falls	Rises	Less CO_2 is fixed, so less RuBP is used and less GP is formed. GP is still being reduced and RuBP is still being regenerated, so RuBP builds up and GP falls.

The model below lets you make these changes and watch the two intermediates respond.

COMMON MISTAKE

Do not say that more light makes the Calvin cycle enzymes work faster. Light affects the light-independent stage only through the supply of ATP and reduced NADP.

10. Quick checks

QUICK CHECK

Why do leaves appear green?

- Chlorophyll absorbs green light most strongly.
- Carotenoids absorb blue and red light and reflect yellow light.
- Chlorophyll absorbs red and blue-violet light and reflects most of the green light.
- The stroma contains a green pigment that absorbs all wavelengths.

Chlorophyll a absorbs most strongly in the blue-violet and red regions, so mostly green light is reflected to the eye. The commonest wrong answer says chlorophyll absorbs green light, which is the reverse of what happens.

QUICK CHECK

Which statement about ATP is correct?

- Phosphorylation of ADP requires energy, and hydrolysis of ATP releases energy.
- Hydrolysis of ADP releases energy, and phosphorylation of ATP requires energy.
- Phosphorylation of ADP releases energy that drives the Calvin cycle.
- ATP contains deoxyribose and three phosphate groups.

Adding Pi to ADP (phosphorylation) needs an energy input; removing it by hydrolysis releases energy for cell processes. The commonest wrong answer swaps the two processes. ATP contains ribose, not deoxyribose.

QUICK CHECK

In non-cyclic photophosphorylation, where do the electrons that replace those lost by photosystem I come from?

- From the photolysis of water.
- From reduced NADP made in the stroma.
- From cyclic flow, which returns them to P700.
- From photosystem II, passed along the electron carriers.

Electrons from photosystem II travel along the carrier chain to photosystem I. Photolysis replaces the electrons lost by photosystem II, not those lost by photosystem I, which is the commonest wrong answer.

QUICK CHECK

The Calvin cycle fixes 6 CO_2 to form one glucose. How many ATP and reduced NADP are used in total?

- 12 ATP and 12 reduced NADP
- 18 ATP and 12 reduced NADP
- 12 ATP and 18 reduced NADP
- 6 ATP and 6 reduced NADP

Each of the 12 GP needs one ATP and one reduced NADP to be reduced to GALP, giving 12 of each. Regenerating 6 RuBP needs another 6 ATP, so the total is 18 ATP and 12 reduced NADP. The commonest wrong answer, 12 of each, leaves out the ATP used to regenerate RuBP.

QUICK CHECK

A pigment travels 4.0 cm while the solvent front travels 10.0 cm. What is its Rf value?

- 0.25
- 2.5
- 6.0
- 0.40

Rf = distance moved by the pigment ÷ distance moved by the solvent = 4.0 ÷ 10.0 = 0.40. The commonest wrong answer, 2.5, comes from dividing the other way round, which gives a value above 1 that is impossible for an Rf.

QUICK CHECK

A plant in bright light is suddenly placed in dim light. What happens to the concentrations of GP and RuBP in its chloroplasts?

- GP falls and RuBP rises.
- GP rises and RuBP falls.
- Both rise.
- Both stay the same.

With less ATP and reduced NADP, GP cannot be reduced to GALP and accumulates, while RuBP is still used for fixation but cannot be regenerated, so it falls. GP falling and RuBP rising is what happens when the carbon dioxide concentration falls, which is the commonest confusion.

QUICK CHECK

What does cyclic photophosphorylation produce?

- Oxygen and ATP
- Reduced NADP only
- ATP only
- ATP, reduced NADP and oxygen

Cyclic flow uses photosystem I only, and the electrons return to P700, so hydrogen ions are pumped and ATP is made, but water is not split and NADP is not reduced. The commonest wrong answer lists the products of non-cyclic flow.

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Ecology, environment and climate change

Ecology asks why organisms live where they do, how many of them there are, and how energy and carbon move between them and their surroundings. This page covers energy flow, distribution and abundance, succession, the carbon cycle, the greenhouse effect, the evidence for climate change and its effects on living things, and how a changing climate drives evolution and the formation of new species.

1. Key terms

- **Habitat:** the place where an organism lives, such as a rocky shore or a rainforest. A microhabitat is a small part of a habitat with its own conditions.
- **Population:** all the individuals of one species living and breeding together in a habitat.
- **Community:** all the populations of all the different species living in a habitat at the same time.
- **Ecosystem:** a community together with its physical and chemical environment, including the nutrients that cycle through it.
- **Niche:** the role of a species in its community: what it feeds on, what feeds on it, where and when it is active, and the conditions it needs.
- **Abiotic factors:** non-living conditions such as temperature, light, water, oxygen and soil pH. **Biotic factors:** effects of other living things, such as competitors, predators, parasites and disease.
- **Abundance and distribution:** abundance is how many individuals of a species are present; distribution is where the species is found.

2. Energy flow through ecosystems

Almost all the energy in an ecosystem comes from the Sun. Producers capture only about 1 to 5% of the light that falls on them. The rest is reflected, passes straight through the leaf, is the wrong wavelength to be absorbed, misses the chloroplasts or is used to evaporate water. The energy that is captured becomes chemical energy in organic molecules and then passes along food chains. Energy is lost from the chain as heat at every step, so it flows through an ecosystem in one direction and must be replaced constantly by new sunlight. Each feeding position in a food chain is a trophic level.

Productivity is the rate at which energy is stored as new biomass, so it is given per unit area per unit time, usually kJ m^{-2} $year^{-1}$. Gross primary productivity (GPP) is the total energy that plants fix in photosynthesis. Plants use part of it in their own respiration (R). The rest, net primary productivity (NPP), is stored in new plant biomass and is the energy available to herbivores and decomposers.

KEY IDEA

NPP = GPP − R, where R is the energy used in respiration by the plants. Only NPP is available to the next

trophic level.

Consumers waste much of the energy in their food. Some plant material is never eaten, such as roots and bark, and some cannot be digested, such as cellulose and lignin, so it leaves as faeces. Energy is also lost in urine, and much is used in respiration for movement and to keep warm. Only what is left builds new body tissue. This is the net secondary productivity of the consumer: energy in food eaten − energy in faeces and urine − energy used in respiration. Dead organisms, faeces, shed fur and skin all pass energy to decomposers.

KEY IDEA

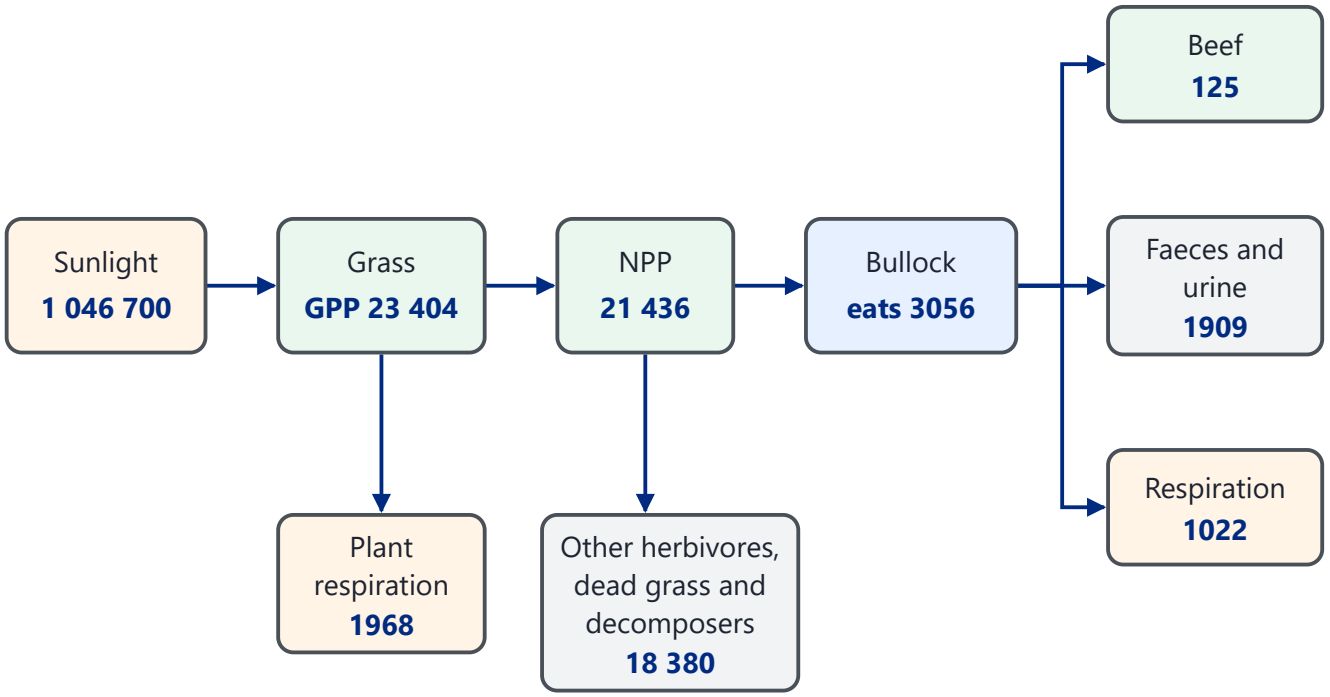
Efficiency of energy transfer (%) = (energy available at one trophic level ÷ energy available at the previous trophic level) × 100. Transfers are usually inefficient, often about 10% and sometimes much less, which is why food chains rarely have more than four or five levels.

Efficiency of transfer from grass to insects

1. A salt-marsh grass, *Spartina*, has a GPP of 36 000 kJ m^{-2} $year^{-1}$ and loses 28 000 kJ m^{-2} $year^{-1}$ in respiration, so its NPP is 36 000 − 28 000 = 8000 kJ m^{-2} $year^{-1}$.
2. Herbivorous insects take in 300 kJ m^{-2} $year^{-1}$ of this energy.
3. Efficiency = 300 ÷ 8000 × 100 = 3.75%.

The energy of the previous level, here the producers' NPP, goes on the bottom because it is the energy that was available to be transferred.

The grazing food chain below shows the same ideas in one square metre of grassland. Of 1 046 700 kJ m^{-2} $year^{-1}$ of sunlight, the grass fixes 23 404 kJ as GPP, an efficiency of 23 404 ÷ 1 046 700 × 100 = 2.24%. Its NPP is 21 436 kJ, so plant respiration is 23 404 − 21 436 = 1968 kJ. A bullock eats 3056 kJ, loses 1909 kJ in faeces and urine and 1022 kJ in respiration, and stores 125 kJ as beef: 125 ÷ 3056 × 100 = 4.1% of the food it eats. As a share of the original sunlight, the beef is 125 ÷ 1 046 700 × 100 = 0.012%.



All values in kJ m^{-2} $year^{-1}$

A grazing food chain (kJ m^{-2} $year^{-1}$): most energy leaves as respiration, faeces or dead material rather than as beef.

Transfer from producers to herbivores is usually less efficient than transfer between carnivores. Plants contain a lot of cellulose and lignin that animals cannot digest, much plant material is inedible and passes to decomposers, and plant tissue holds less energy per gram than animal tissue, which is mostly protein and fat.

EXAM TIP

If asked why keeping cattle indoors raises the efficiency of energy transfer, explain the cause: they move less and lose less heat keeping warm, so less energy is used in respiration and more of their food energy becomes new body mass.

What changes productivity

NPP varies from place to place. More light gives more photosynthesis, which is why NPP falls with depth in water. Warmer temperatures speed up the enzymes of photosynthesis, up to the optimum. More rainfall provides water for photolysis and for carrying mineral ions, so GPP and NPP rise. Low carbon dioxide or a shortage of nitrate and phosphate also limit growth. In an old forest, NPP falls because the trees grow more slowly and shade one another, and more of the energy fixed is used to maintain the large mass of trunks and branches. NPP is averaged over a whole year because it changes with the seasons.

3. Distribution and abundance

Each species can only survive and reproduce within a range of each condition, its range of tolerance, and it does best near the optimum. The numbers and distribution of organisms in a habitat are therefore controlled by abiotic factors and by biotic factors.

Abiotic factors

Temperature acts mainly through enzymes. Below 0 °C, water in cells freezes, ice crystals damage cells, and enzymes have too little kinetic energy to work. Above about 40 °C, most enzymes and other proteins start to denature, so metabolism fails. Light provides the energy for photosynthesis, so it controls primary productivity and, indirectly, the food supply of every other organism. Some plants are adapted to shade, for example with large leaves or more chlorophyll, and seasonal changes in day length control breeding and behaviour in many animals.

Water availability depends on rainfall, evaporation and drainage, and controls the germination and growth of plants. In water, oxygen is often limiting: cold, fast-flowing water holds more dissolved oxygen than warm, still water. Waterlogged soil has little oxygen for root respiration, so only specially adapted plants grow there. Wind increases cooling and water loss, and strong currents or storms can remove whole populations. Soil (edaphic) factors include pH, which affects enzymes and the availability of nutrients, and the mineral content and structure of the soil. Topography matters too: in the northern hemisphere a south-facing slope is warmer because it receives more sun, steep slopes drain quickly, and higher altitude is colder and windier.

Biome	Typical conditions	Biodiversity
Tropical rainforest	Warm and wet all year, plenty of light	Very high
Temperate forest	Mild, moist, distinct seasons	Medium to high
Temperate grassland	Warm summers, cold winters, low rainfall	Medium
Taiga (coniferous forest)	Long cold winters, short summers	Low
Tundra and desert	Very cold, or very dry with extreme temperatures	Very low

Biotic factors

Competition happens when organisms need the same limited resource, such as food, water, light, space, mates or nesting sites. Intraspecific competition is between members of the same species. When resources are scarce, fewer individuals survive and breed, so population growth slows; when resources are plentiful, numbers rise. Interspecific competition is between different species whose niches overlap. It reduces the abundance of both, and if the overlap is large one species can eventually replace the other completely. Experiments with two species of *Paramecium* show this: each grows well alone, but grown together one dies out.

A predator kills and eats living prey. Predator and prey numbers often rise and fall in cycles. When prey is plentiful, more predators survive and breed; they then eat prey faster than it reproduces, so prey numbers fall; with less food, predator numbers fall, and the prey can recover. The predator peak comes after the prey peak. Parasites live on or in a host and harm it, and disease weakens individuals so they breed less and are caught more easily. In mutualism both partners benefit.

Density-dependent and density-independent factors

Density-independent factors affect a population whatever its size. Extremes of temperature, floods and storms are examples; they mainly limit where a species can live. Density-dependent factors have a greater effect when a population is crowded. Disease and parasites spread more easily, competition for food and territory is stronger, and individuals without a territory are less able to breed. These factors mainly control abundance.

How the niche explains distribution and abundance

Two species cannot share exactly the same niche in the same habitat for long, because the better competitor will outcompete the other. Species that live together therefore differ in at least some part of their niche, such as their food, the part of the habitat they use or the time they are active. A habitat with more niches, such as a mixed woodland with several layers of vegetation, can support more species. Some insects avoid competition between their own young and adults because the larvae and adults feed on different foods and so occupy different niches.

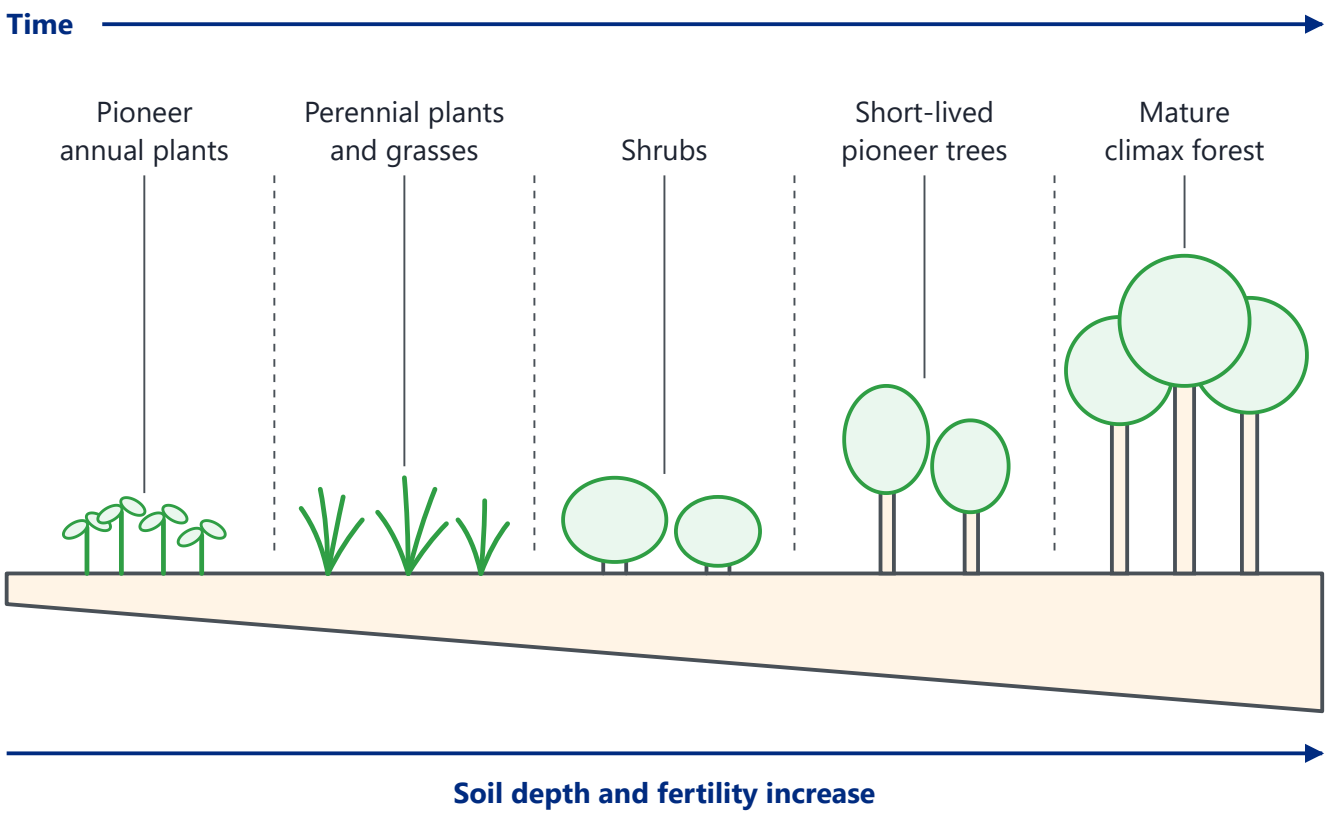
To measure abundance, ecologists sample the habitat. For plants, quadrats are placed at random coordinates read along two tape measures, so that the choice of position is not biased.

4. Succession

Succession is the change in the communities living in an area over time. Each community changes the abiotic conditions, for example by adding organic matter to soil or by shading the ground, and this allows other species to colonise and outcompete the earlier ones. The process continues until a stable climax community is reached.

Primary succession

Primary succession starts on bare, lifeless ground with no soil, such as new volcanic rock or rock left by a retreating glacier. The conditions are harsh: there is no soil to hold water or minerals. The first colonisers, pioneer species such as lichens and mosses, are adapted to survive there. They help to break the rock into small grains, and when they die, decomposers break down their remains into humus. Rock grains and humus together form a thin soil. Plants with small, shallow roots, such as grasses, can now grow; as they die they add more organic matter, so the soil becomes deeper, holds more water and minerals and becomes more fertile. Larger plants follow: shrubs, then fast-growing pioneer trees, and finally the slower-growing trees of the climax forest.



As soil depth and fertility increase, each stage is replaced by larger plants until a climax forest forms.

Through a succession, the number of species and the total biomass increase, more organic matter is decomposed, and the community's total respiration rises. Because respiration catches up with photosynthesis, net community production falls as the community matures.

Secondary succession

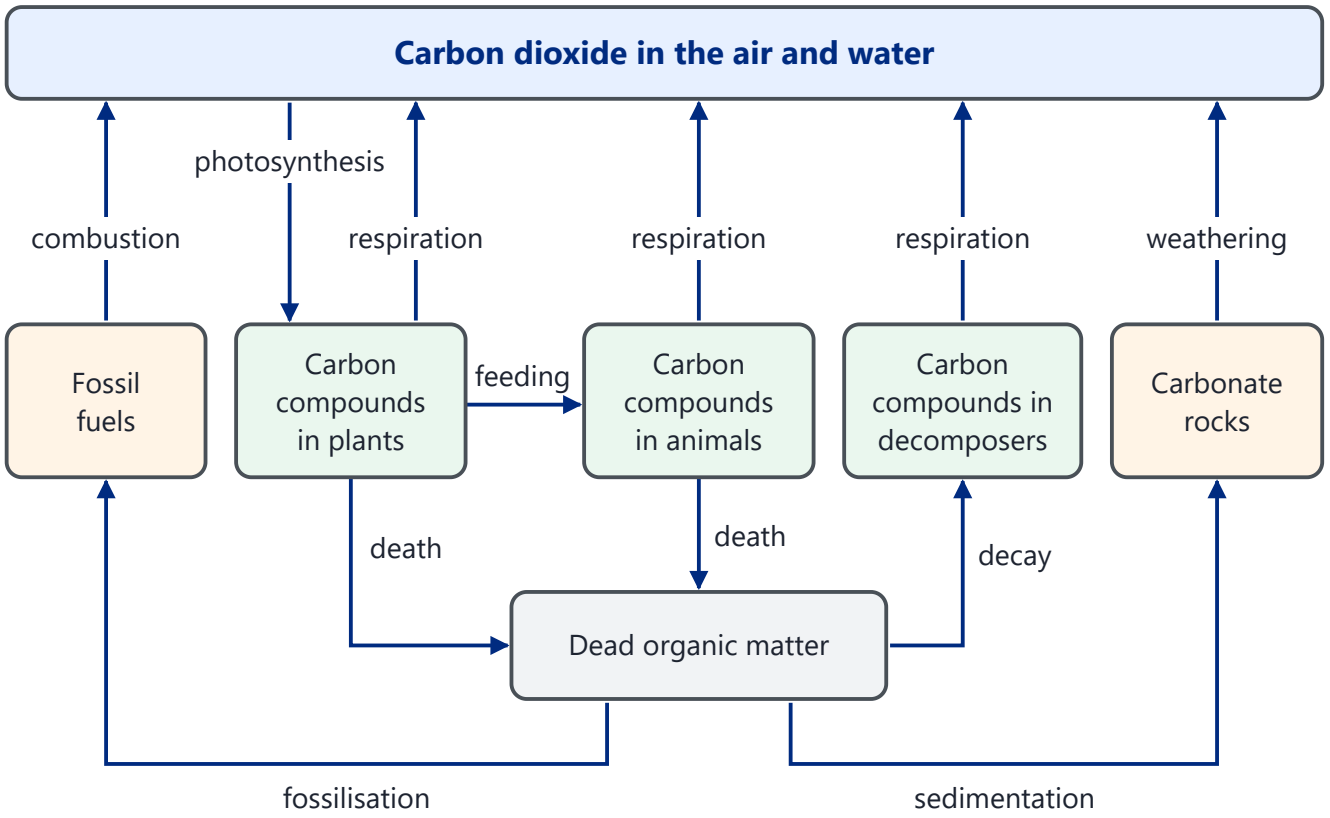
Secondary succession happens where soil is already present but the vegetation has been removed, for example after a fire or after a forest has been cleared. The soil contains seeds, roots and soil organisms, so plants grow quickly as soon as there is light and water. The sequence is similar to primary succession, but it is faster and starts with more species.

Climax communities and plagioclimax

A climax community is stable and complex, with many species and many interactions, and it stays the same unless the environment changes. Which climax forms depends on the climate, so it is called a climatic climax. In a temperate climate it is usually forest, because once deep soil has formed there is enough water and the temperatures are mild enough for large trees. In a polar climate the climax has only herbs or low shrubs, because water is scarce, temperatures are low and the seasons are extreme. Human activities such as mowing, grazing and burning can stop succession at an earlier stage. The community that results is a plagioclimax: a regularly mown field stays as grassland even though the climate could support woodland.

5. The carbon cycle

Carbon cycles between the atmosphere, living organisms, dead organic matter, the oceans and rocks. Plants take in carbon dioxide in photosynthesis and build it into carbohydrates, proteins and lipids. Carbon passes to animals when they feed. All organisms return carbon dioxide to the air in respiration. Where dead material cannot decay, for example in waterlogged bogs, carbon can be stored for millions of years and may form fossil fuels. Burning fuels releases this carbon again.



The carbon cycle: rapid exchanges between organisms and the air, and slow storage in fossil fuels and rock.

The role of microorganisms

Decomposers, mainly bacteria and fungi, secrete enzymes onto dead organic matter. The enzymes hydrolyse large molecules, such as starch, cellulose and proteins, into small soluble ones, for example by breaking glycosidic bonds to release glucose. The microorganisms absorb these products and use them in respiration, releasing carbon dioxide that plants can fix again. Without decomposers, carbon would stay locked in dead material and the supply of carbon dioxide for photosynthesis would run down. Decomposition also returns minerals such as nitrate to the soil.

Carbon sinks

A carbon sink is a store that holds carbon out of the atmosphere for a long time. Limestone and other carbonate rocks hold the largest store. The oceans hold a great deal of dissolved carbon and carbon in the shells of marine

organisms. Fossil fuels, peat and forests are also major stores. Carbon in these stores rarely escapes unless the store is burned, drained, cleared or broken down.

How humans increase atmospheric carbon dioxide

Burning fossil fuels releases carbon that was locked away for millions of years, so it adds to the amount in the air. Deforestation raises carbon dioxide in two ways: fewer trees are left to remove it by photosynthesis, and the carbon in the wood is released when it is burned or decays. Clearing forest also warms the bare soil, which speeds up respiration by soil microorganisms and the breakdown of soil organic matter. Accurate measurements since 1957 show a rising trend, from about 315 ppm in 1957 to about 350 ppm by 1988, with a yearly cycle: the level is lower in summer, when photosynthesis is fastest, and higher in winter.

Reducing atmospheric carbon dioxide

There are two approaches: put less carbon dioxide into the air, by burning less fossil fuel and using energy more efficiently, or take more out of it, by increasing photosynthesis. Reforestation means planting young, fast-growing trees. While they grow, they photosynthesise faster than they respire, so they remove carbon dioxide and lock carbon in their wood. Its limits are that a mature forest is roughly carbon neutral, because its photosynthesis is balanced by respiration and decay, and that there is only a limited amount of land to plant.

Biofuels are fuels made from recently living material, often crops that can be replanted. Burning them releases only the carbon dioxide the crop took in as it grew, so in principle they are carbon neutral. In practice, carbon dioxide is released when land is cleared for the crop and by the machinery used to grow, harvest and process it, so there may be a net increase. Ploughing less also helps: less air enters the soil, so aerobic decomposers break down organic matter more slowly and more carbon stays in the soil.

Biofuels: advantages	Biofuels: disadvantages
Reduce the use of limited fossil fuels	Clearing forests to grow them destroys carbon sinks and releases stored carbon
Renewable, and can reduce net carbon emissions	Loss of habitat and biodiversity
Waste biomass can be used to make them	Food shortages where fuel crops replace food crops

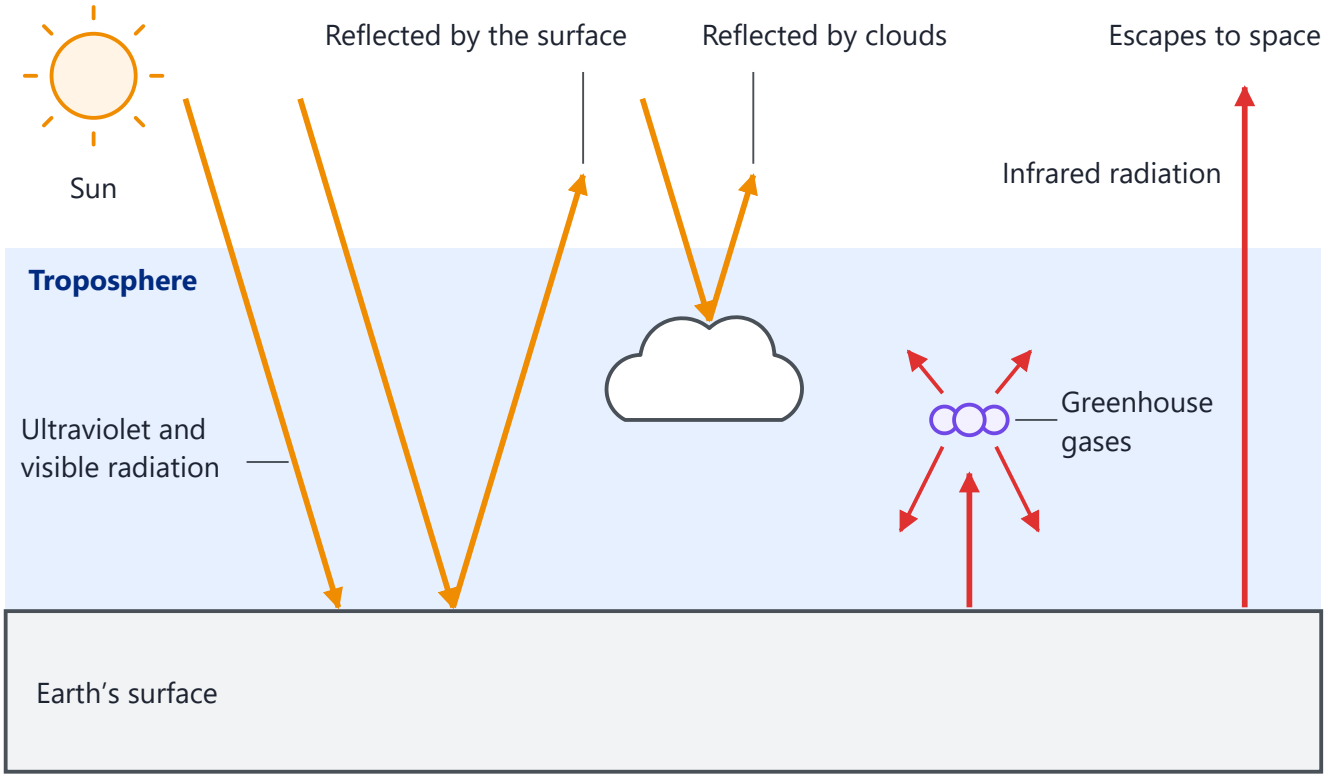
KEY IDEA

Reforestation and the sustainable use of resources such as biofuels are examples of managing the conflict between human needs, for land, food, timber and energy, and conservation. Good management meets people's needs while protecting habitats and carbon stores.

6. The greenhouse effect

Energy from the Sun reaches Earth mainly as short-wavelength visible and ultraviolet radiation. Some is reflected by clouds and the surface, and most of the rest is absorbed by the surface, which warms and then emits longer-wavelength infrared radiation. Greenhouse gases, such as carbon dioxide, methane and water vapour, absorb

some of this infrared and re-emit it in all directions, including back towards the surface, so the lower atmosphere stays warmer. This natural greenhouse effect keeps Earth warm enough for life.



Greenhouse gases absorb infrared radiation emitted by Earth's surface and re-emit it in all directions, some of it back towards the surface, so the lower atmosphere stays warmer.

Human activities are increasing the concentrations of greenhouse gases, which enhances the greenhouse effect. Less infrared escapes to space, so the mean surface temperature rises. This is global warming, and the human-caused part of climate change is called anthropogenic climate change. Carbon dioxide comes from burning fossil fuels and from deforestation. Methane comes from anaerobic microorganisms in landfill sites, paddy fields and the guts of cattle and other ruminants, and from extracting fossil fuels. As the world warms, frozen ground (permafrost) may thaw and release stored methane, which would add to the warming.

- **Weather:** the conditions in the atmosphere at a particular place and time.
- **Climate:** the average weather of a large area over a long period.
- **Climate change:** a large-scale, long-term change in global or regional weather patterns.

7. Evidence for climate change

Several independent kinds of evidence show that the climate is changing. Each has strengths and weaknesses, which is why scientists use them together.

Temperature and carbon dioxide records

The Central England Temperature record, which begins in 1659, is the longest continuous instrumental temperature record in the world. It shows large fluctuations from year to year, but an overall rise, which has been steepest over the last century. Direct measurements of atmospheric carbon dioxide since 1957 show a steady rise. Instrument records are accurate, but they only cover the last few centuries, so other evidence is needed for earlier times.

Tree rings (dendrochronology)

In seasonal climates a tree adds one growth ring each year, so counting the rings gives the age of each ring, and the width of a ring shows how much the tree grew that year. Warmer conditions speed up the enzymes of photosynthesis, so more material is laid down and the ring is wider; very high temperatures above the optimum would reduce growth. A core from a tree whose recent rings are wider than older ones suggests the climate has become warmer. By matching ring patterns between living trees and older dead or preserved wood, scientists build records that go back thousands of years. Ring width also depends on rainfall, sunlight and carbon dioxide, so it is hard to be sure which factor caused a wide ring, and an unusual year can produce more than one ring.

Pollen in peat bogs

Peat bogs are acidic, waterlogged and low in oxygen, so decomposers cannot work well and pollen is preserved. Peat builds up in layers, so the deeper the layer, the older it is. Pollen grains have a tough outer wall and a shape that identifies the plant type. Scientists take cores, identify the pollen in each layer, and use the climate in which those plants grow today to infer the climate when the layer formed. For example, if oak pollen increases while pine pollen decreases, the climate was probably becoming warmer, because oak outcompetes pine in warmer conditions. The record only shows plants that were successful and produced plenty of pollen at the time.

Ice cores

Snow falling on ice sheets is compressed into ice in yearly layers that can be counted. Bubbles of air trapped in each layer are samples of the atmosphere at that time, so their carbon dioxide and methane can be measured, and the ratio of oxygen isotopes in the ice indicates the temperature when the snow fell. Ice cores give records going back hundreds of thousands of years.

Correlation, causation and models

Between 1950 and the early 2000s, both mean global temperature and atmospheric carbon dioxide rose, so there is a positive correlation between them. A correlation alone does not show that one variable causes the other, because other factors, such as changes in solar activity, might be involved. A causal link needs a mechanism, here the absorption of infrared by carbon dioxide, and further studies that rule out or control the other factors.

Data collected so far can be extrapolated, by extending the trend, to predict future change, and these predictions are used to build computer models of the future climate. Models have limitations: the data are incomplete, understanding of how the climate system works is incomplete, computer power is limited, not every factor can be included, and nobody can know how human use of energy and resources will change. Extrapolations that ignore year-to-year fluctuations can also be misleading.

8. Effects of climate change on living things

Temperature, enzymes and Q10

Most effects of warming on organisms come back to enzymes. Up to the optimum, a rise in temperature increases kinetic energy, so enzyme and substrate molecules collide more often and more enzyme-substrate complexes form. Metabolism speeds up, so organisms grow faster and pass through their life cycles more quickly. Above the optimum, enzymes denature, metabolism slows, growth slows and the organism may die. This applies to plants, animals and microorganisms.

The temperature coefficient, Q10, measures how much the rate of a reaction changes for a 10 °C rise in temperature:

KEY IDEA

Q10 = rate of reaction at (T + 10) °C ÷ rate of reaction at T °C. For most enzyme-controlled reactions between about 0 °C and 40 °C, Q10 is about 2, so the rate roughly doubles for every 10 °C rise. A Q10 of 3 would mean the rate trebles.

Calculating Q10

1. Read the rate at two temperatures 10 °C apart, below the optimum. Suppose the rate is 12 units at 15 °C and 27 units at 25 °C.
2. Divide the rate at the higher temperature by the rate at the lower one: Q10 = 27 ÷ 12 = 2.25.

Q10 only describes the rising part of the curve. Above the optimum, denaturation takes over and the rate falls.

CORE PRACTICAL

Core practical: the effect of temperature on development. Hatch brine shrimp eggs, or grow seedlings, at a range of controlled temperatures, keeping other variables such as salinity, light and the number of eggs or seeds the same. Record the percentage hatched or the growth over time. Development speeds up as temperature rises towards the optimum and slows above it.

Distribution of species

Every species lives where its conditions suit it. As the climate changes, those conditions move. Animals may migrate to places that now suit them, which can increase biodiversity there but brings new competition with the species already present. Plants spread far more slowly and may not keep up. Species that cannot move or adapt

may die out locally, while others extend their range into areas that used to be too cold; for example, a warm-climate crop pest could spread further from the equator. Changing rainfall patterns matter too: some regions will become wetter and others drier, deserts may expand, and species not adapted to dry conditions must move or die out. Warming may also let disease vectors such as mosquitoes spread into new areas.

Development and life cycles

Many organisms time their life cycles to seasonal cues such as temperature, rainfall and food supply. Warming brings spring earlier, so plants flower earlier and insects emerge earlier. Animals that rely on these food sources may be thrown out of step. Great tits in the Netherlands now breed earlier each year, but the caterpillars they feed to their chicks are emerging even earlier, so many chicks miss the food peak and fewer survive. Migrating birds, such as swallows arriving from Africa at their usual time, may find that the insect peak has already passed. Changes in plants then affect herbivores and the predators that eat them.

Sex ratios and decomposition

In some reptiles, the incubation temperature of the eggs decides the sex of the young. In some crocodiles, males develop only in a narrow temperature range of about 32 to 33 °C, and cooler or warmer eggs become females. A warmer climate could produce few or no males, and the population could die out.

Warmer soil also speeds up the enzymes of decomposers, so dead organic matter is broken down faster and more carbon dioxide and methane are released by their respiration. This adds more greenhouse gases and could increase warming further.

9. Climate change and evolution

Climate change as a selection pressure

As the climate becomes warmer, colder, wetter or drier in different places, organisms face new selection pressures. Evolution is a change in the frequency of alleles in a population over time, and natural selection is one way it happens. A rise in temperature does not cause the mutations that natural selection acts on. Mutations happen at random, all the time; the change in the environment only changes which alleles give an advantage.

How natural selection changes allele frequency

1. Gene mutations produce new alleles, so individuals in a population vary in their alleles and phenotypes.
2. A selection pressure, such as a warmer climate, means that some individuals are better adapted than others.
3. Individuals with an advantageous allele are more likely to survive, reproduce and pass that allele on to their offspring.

Natural selection acts on phenotypes, but only the alleles are inherited.

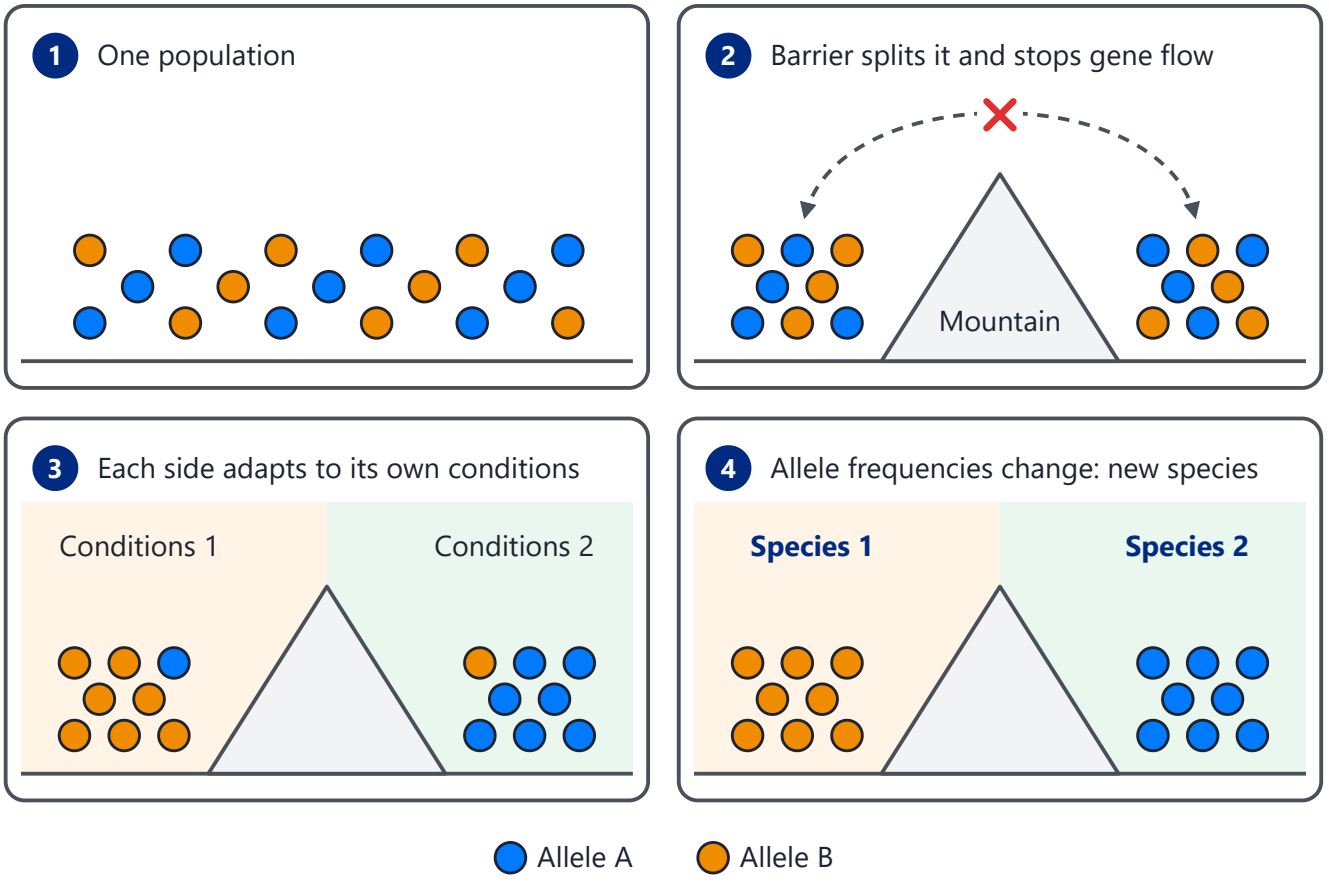
4. A greater proportion of the next generation carries the advantageous allele, and the process repeats each generation.

5. The frequency of the advantageous allele increases over generations. This change in allele frequency is evolution.

Isolation and the formation of new species

A species is a group of organisms that can interbreed to produce fertile offspring. A new species forms when two populations of one species become reproductively isolated. Isolation stops gene flow, the movement of alleles between the populations, so mutations, natural selection and chance change allele frequencies separately in each. Eventually the populations become so genetically different that they could no longer interbreed to produce fertile offspring even if they met. This is speciation.

In allopatric speciation the populations are separated geographically, for example by a mountain range, a river or the sea. The conditions on each side differ, such as the climate, so each population experiences different selection pressures. Different alleles become common in each gene pool, and over many generations the populations become separate species.



Allopatric speciation: a physical barrier stops gene flow, the two populations adapt to different conditions, and they become separate species.

In sympatric speciation there is no geographical barrier: the populations live in the same area. Reproductive isolation arises because random mutations change the phenotype of some individuals in a way that stops them breeding with the rest. For example, if some insects in a population start to feed and mate on a different food plant, they will tend to mate mainly with each other. Gene flow between the groups falls, the groups diverge, and in time a new species may form.

Reproductive isolation can work in several ways. Some mechanisms stop mating or fertilisation from happening; others stop a hybrid from developing or from breeding.

Mechanism	What happens
Seasonal (temporal)	Groups flower or become ready to mate at different times of year.
Behavioural	Groups develop different courtship displays or songs that do not attract the other group.
Mechanical	Changes in the reproductive organs prevent successful mating.
Ecological (habitat)	Groups live in the same area but use different habitats, so they rarely meet.
Hybrid inviability	A hybrid embryo forms but fails to develop or dies young.
Hybrid sterility	The hybrid reaches adulthood but cannot produce gametes. A mule, from a horse (2n = 64) and a donkey (2n = 62), has 63 chromosomes that cannot pair in meiosis.

Evidence and the scientific community

The theory of evolution by natural selection is supported by many kinds of evidence, and molecular biology has added more. Comparing DNA base sequences, and the amino acid sequences of proteins (proteomics), shows how closely species are related: the more recently two species shared a common ancestor, the more similar their sequences are.

New evidence only becomes accepted after it has been checked by other scientists. Researchers publish their methods and results in scientific journals. Before publication, the paper goes through peer review: other experts in the field check that the methods are valid, that the conclusions follow from the data and that the work is original. Scientists also present findings at scientific conferences, where others can question them. Because methods are published, other groups can repeat the work and confirm or challenge the results. This process makes it much less likely that errors, poor methods or false claims are accepted.

Conclusions about controversial issues, such as how far humans are causing climate change or what should be done about it, can depend on who is interpreting the evidence and how it is presented. Some argue that cutting fossil fuel use alone will not stop warming, because carbon dioxide also comes from decomposition and deforestation, other gases such as methane contribute, and natural events such as volcanic eruptions can affect climate. People may select the data that suit their view, a source may be biased by who funds it, and the media may simplify or exaggerate. When judging a claim, ask who produced it, whether it has been peer reviewed, and whether it is supported by other independent evidence.

10. Using statistics with ecological data

Ecological and climate data vary a great deal, so a statistical test is used to decide whether a pattern is real or could have happened by chance. Each test starts with a null hypothesis, which states that there is no correlation or no difference. The calculated value is compared with a critical value from a table, usually at the 5% significance level ($p = 0.05$). If the calculated value is greater than or equal to the critical value, the null hypothesis is rejected: there is less than a 5% probability that the result is due to chance.

- **Spearman's rank correlation coefficient (r_s):** tests whether there is a significant correlation between two variables, such as temperature and tree ring width. $r_s = 1 - 6\Sigma d^2 \div n(n^2 - 1)$, where d is the difference between the ranks of each pair and n is the number of pairs. It runs from +1 (perfect positive correlation) to −1 (perfect negative correlation).
- **Student's t-test:** tests whether the difference between the means of two sets of data is significant, such as hatch rates at two temperatures. $t = (\bar{x}_1 - \bar{x}_2) \div \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$, where s is the standard deviation, and degrees of freedom $= n_1 + n_2 - 2$.

Year	Mean temperature / °C	Rank	Ring width / mm	Rank	d	d²
1	9.1	2	1.2	1	1	1
2	9.4	3	1.5	3	0	0
3	9.0	1	1.3	2	−1	1
4	9.8	4	1.6	4	0	0
5	10.2	6	1.9	6	0	0
6	9.9	5	1.8	5	0	0
7	10.5	8	2.0	7	1	1
8	10.4	7	2.2	8	−1	1

Spearman's rank for temperature and ring width

1. Null hypothesis: there is no correlation between mean temperature and ring width.
2. Rank each variable separately from lowest to highest, find the difference in rank (d) for each year, square it and add: $\Sigma d^2 = 4$.
3. $r_s = 1 - (6 \times 4) \div (8 \times (64 - 1)) = 1 - 24 \div 504 = 1 - 0.048 = 0.95$.
4. The critical value for 8 pairs at $p = 0.05$ is 0.738. Since 0.95 is greater than 0.738, reject the null hypothesis: there is a significant positive correlation.

A significant correlation still does not prove that the higher temperature caused the wider rings; rainfall or carbon dioxide could have changed too.

11. Quick checks

QUICK CHECK

A plant fixes 23 404 kJ m^{-2} $year^{-1}$ as GPP and uses 1968 kJ m^{-2} $year^{-1}$ in respiration. What is its NPP?

- 25 372 kJ m^{-2} $year^{-1}$
- 1968 kJ m^{-2} $year^{-1}$
- 21 436 kJ m^{-2} $year^{-1}$
- 23 404 kJ m^{-2} $year^{-1}$

NPP = GPP − R = 23 404 − 1968 = 21 436 kJ m^{-2} $year^{-1}$. The commonest wrong answer adds respiration to GPP instead of subtracting it.

QUICK CHECK

Why is the transfer of energy from producers to herbivores usually less efficient than from herbivores to carnivores?

- Plants contain much indigestible cellulose and lignin, and much plant material is not eaten.
- Herbivores respire less than carnivores.
- Energy is destroyed when plants are eaten.
- Plants store more energy per gram than animal tissue.

Herbivores cannot digest cellulose and lignin, so much of their food leaves as faeces, and roots and woody parts are often not eaten at all. Energy is never destroyed; it is lost as heat or passed to decomposers. Plant tissue actually holds less energy per gram than animal tissue.

QUICK CHECK

In a grassland kept by regular mowing, the community does not progress to woodland. What is this community called?

- A climatic climax
- A pioneer community
- A primary succession
- A plagioclimax

A plagioclimax is a community held at an earlier stage by human activity, here mowing. A climatic climax is the stable community that the climate would support without interference, which is the commonest confusion.

QUICK CHECK

Why is pollen preserved in peat bogs?

- The bogs are warm, so pollen dries out quickly.
- Acidic, waterlogged, low-oxygen conditions slow the activity of decomposers.
- Pollen contains no organic molecules.
- Decomposers cannot reach the surface of the bog.

Low pH inhibits decomposers' enzymes, and lack of oxygen limits their aerobic respiration, so the tough pollen grains are not broken down. Pollen is organic, so it would decay under normal conditions.

QUICK CHECK

The rate of a reaction is 6 units at 20 °C and 15 units at 30 °C. What is the Q10?

- 0.4
- 9
- 2.5
- 1.5

Q10 = rate at the higher temperature ÷ rate at the lower temperature = 15 ÷ 6 = 2.5. Dividing the other way round gives 0.4, and subtracting gives 9; neither is a Q10.

QUICK CHECK

Two populations of a beetle are separated by a new river. Over many generations they become unable to interbreed. What has happened?

- Sympatric speciation, because both populations are in the same region
- Allopatric speciation, because a geographical barrier stopped gene flow
- No speciation, because the beetles still look alike
- The river caused mutations that made the beetles different

A geographical barrier stopped gene flow, so selection and mutation changed allele frequencies separately until the populations were reproductively isolated. Sympatric speciation happens without a geographical barrier. The environment does not cause the mutations; it selects among them.

QUICK CHECK

Why does a rising carbon dioxide level alone not prove that it caused a rising temperature?

- Carbon dioxide does not absorb infrared radiation.
- The two variables always change in opposite directions.
- Temperature records are never reliable.
- A correlation does not show cause, and other factors such as solar activity may be involved.

A correlation is evidence, but a causal link needs a mechanism and studies that rule out other factors. Carbon dioxide does absorb infrared, which is the mechanism that supports the causal link, so the first option is wrong.

QUICK CHECK

What is the main purpose of peer review?

- Other experts check the methods, data and conclusions of a paper before it is published.
- The authors repeat their own experiment several times.
- The public votes on whether a theory is correct.
- It guarantees that a published conclusion can never be changed.

Peer review is evaluation by independent experts in the same field before publication. It reduces errors but does not make a conclusion permanent; new evidence can still change it.

A2 · Unit 4 · WBI14 Energy, Environment, Microbiology and Immunity

Microbiology and immunity

Most microorganisms are harmless or useful, but some cause disease. This page covers how bacteria and viruses are built and reproduce, how microbiologists culture and count them safely, how tuberculosis and HIV cause disease, how the body defends itself, and how antibiotics work and why bacteria become resistant to them.

1. Bacteria and viruses

A bacterium is a prokaryotic cell. Its DNA is a single circular molecule lying free in the cytoplasm. All bacteria have a cell surface membrane, 70S ribosomes and a peptidoglycan (murein) cell wall, which stops the cell bursting when water enters by osmosis. Some also have plasmids, small loops of DNA that often carry antibiotic resistance genes; flagella for movement; pili, which help cells stick together and can transfer DNA; and a capsule or slime layer that protects them from immune cells.

A virus is not a cell. It is a nucleic acid, DNA or RNA, inside a protein coat (capsid), and some viruses also have a lipid envelope taken from the last host cell, carrying glycoproteins that bind to receptors on new host cells. With no ribosomes or metabolism of its own, a virus can only reproduce inside a living host cell: it is an obligate intracellular parasite.

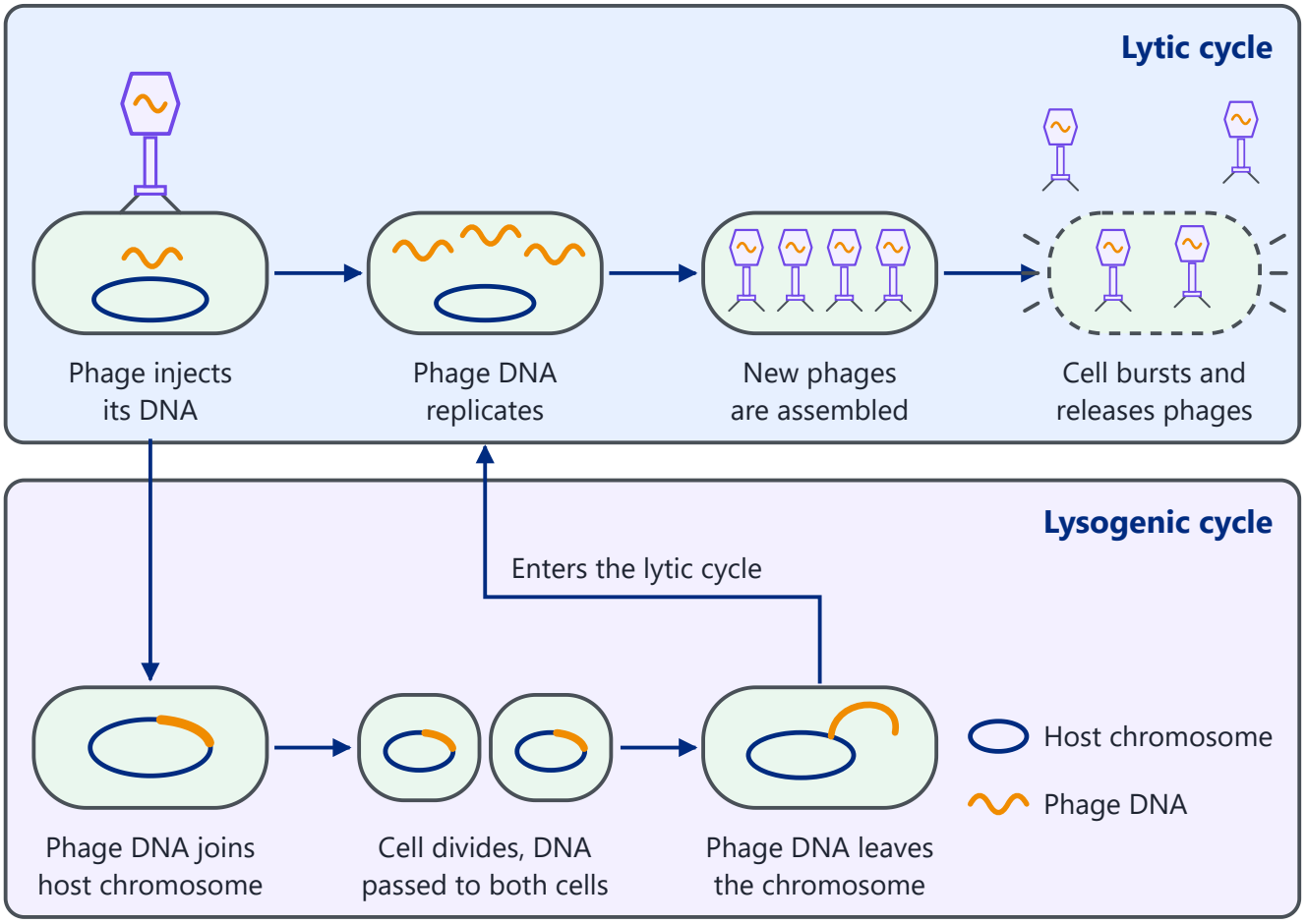
Virus	Nucleic acid	Capsid and envelope	Host
Human immunodeficiency virus (HIV)	Two copies of single-stranded RNA, with the enzyme reverse transcriptase	Capsid inside a lipid envelope carrying glycoprotein gp120	Human helper T cells
Ebola virus	Single-stranded RNA, with an RNA polymerase that copies it	Long, thread-like capsid inside a lipid envelope with glycoprotein spikes	Humans and other primates
Tobacco mosaic virus (TMV)	Single-stranded RNA	Rod-shaped capsid of protein units arranged in a spiral; no envelope	Tobacco and related plants
Lambda (λ) phage	Double-stranded DNA, which forms a circle inside the host	Head (capsid), tail and tail fibres; no envelope	The bacterium *E. coli*

Feature	Bacterium	Virus
Cellular?	Yes, a prokaryotic cell	No, a particle
Nucleic acid	Double-stranded circular DNA, plus any plasmids	DNA or RNA, single- or double-stranded
Outer layers	Membrane and peptidoglycan wall, sometimes a capsule	Protein capsid, sometimes a lipid envelope
Ribosomes and metabolism	Own 70S ribosomes and enzymes	None; uses the host cell's
Reproduction	Binary fission	Only inside a host cell

Lytic and latent infection

Inside a host cell a virus can follow one of two routes. In the lytic pathway, the host's enzymes and ribosomes copy the viral nucleic acid and make capsid proteins, new particles are assembled, and the cell bursts (lyses), releasing them. A virus in this pathway is virulent: it causes disease.

In the lysogenic pathway, the viral DNA is inserted into the host's DNA as a provirus (a prophage in a bacterium) and is copied whenever the host divides. A viral repressor protein stops the other viral genes being transcribed, so the virus is latent: present but inactive. If the host cell is damaged or stressed, the repressor is lost and the virus enters the lytic pathway. HIV and TB can both stay latent for years.



The lytic cycle (top) releases new phages by bursting the host cell. In the lysogenic cycle (bottom) the phage DNA is copied with the host chromosome until it re-enters the lytic cycle.

2. Culturing microorganisms safely

Even a harmless culture carries risks: it could be contaminated by pathogens from the environment, or a pathogenic mutant could arise. Aseptic technique keeps the culture pure and protects the worker and the environment.

- **Sterilise beforehand:** media and glassware are heated in an autoclave at 121 °C for 15 minutes under pressure, which kills bacteria and their spores. Wire inoculating loops are heated in a Bunsen flame until red hot.

- **Work aseptically:** disinfect the bench, work close to a Bunsen flame so that rising air carries microbes away, flame the necks of bottles, and lift Petri dish lids as little as possible.
- **Incubate safely:** tape the lid of a Petri dish without sealing it completely, so that anaerobic pathogens are not encouraged, and incubate at about 25 °C, well below human body temperature, so that human pathogens are not favoured.
- **Dispose safely:** autoclave all cultures and equipment before disposal.

Cultures are grown in nutrient broth (liquid) or on nutrient agar (solid), which supply carbon, nitrogen and minerals; on agar each cell grows into a separate colony. A selective medium lets only certain microorganisms grow: a low-pH medium suppresses bacteria but lets fungi grow, and a medium containing an antibiotic picks out resistant strains, including genetically modified microorganisms carrying a resistance marker gene. Growing cultures with or without oxygen also selects aerobic or anaerobic species. Indicator media change colour around colonies of certain bacteria, so those colonies can be identified.

Growth depends on temperature, pH, nutrients, oxygen and water. Growth is fastest at the optimum temperature of the microorganism's enzymes; above it enzymes denature and cells die. In a fridge (0 to 5 °C) enzymes have little kinetic energy, so growth and decay are slow. In a freezer the cytoplasm freezes, so enzyme and substrate molecules cannot move and collide, and growth stops.

3. Measuring microbial growth

Microbial growth is an increase in the number of cells, or in the mass, of a population.

- **Cell counts:** a haemocytometer is a slide with a chamber of known depth and a ruled grid, so cells in a known volume can be counted under a microscope.
- **Dilution plating:** a culture is serially diluted, a known volume of each dilution is spread on agar, and colonies are counted after incubation. Each colony is assumed to come from one living cell, so this is a viable count. A plate with separate, countable colonies is used, and the count is multiplied by the dilution factor.
- **Turbidity:** a colorimeter measures how cloudy a broth is; more cells absorb more light. A calibration curve, made from samples whose cells have also been counted, converts absorbance into cell numbers.
- **Mass:** cells are filtered or centrifuged from the broth and dried to constant mass. This suits fungi, whose hyphae cannot be counted; fungal growth on agar can also be followed by measuring colony diameter.

Counting cells with a haemocytometer

1. Count the cells in a square of known volume. Suppose there are 15 cells in 0.004 mm^3 of a culture that was diluted 1 in 1000.

   Count cells touching the top and left lines of a square but not those touching the bottom and right lines, so no cell is counted twice.

2. Cells per mm^3 = 15 ÷ 0.004 = 3750.

3. Convert to cm^3 (1 cm^3 = 1000 mm^3): 3750 × 1000 = 3.75 × 10^6 cells per cm^3.

4. Multiply by the dilution factor: $3.75 \times 10^6 \times 1000 = 3.75 \times 10^9$ cells per cm^3 of the original culture.

The count was made on a diluted sample, so the dilution must be reversed.

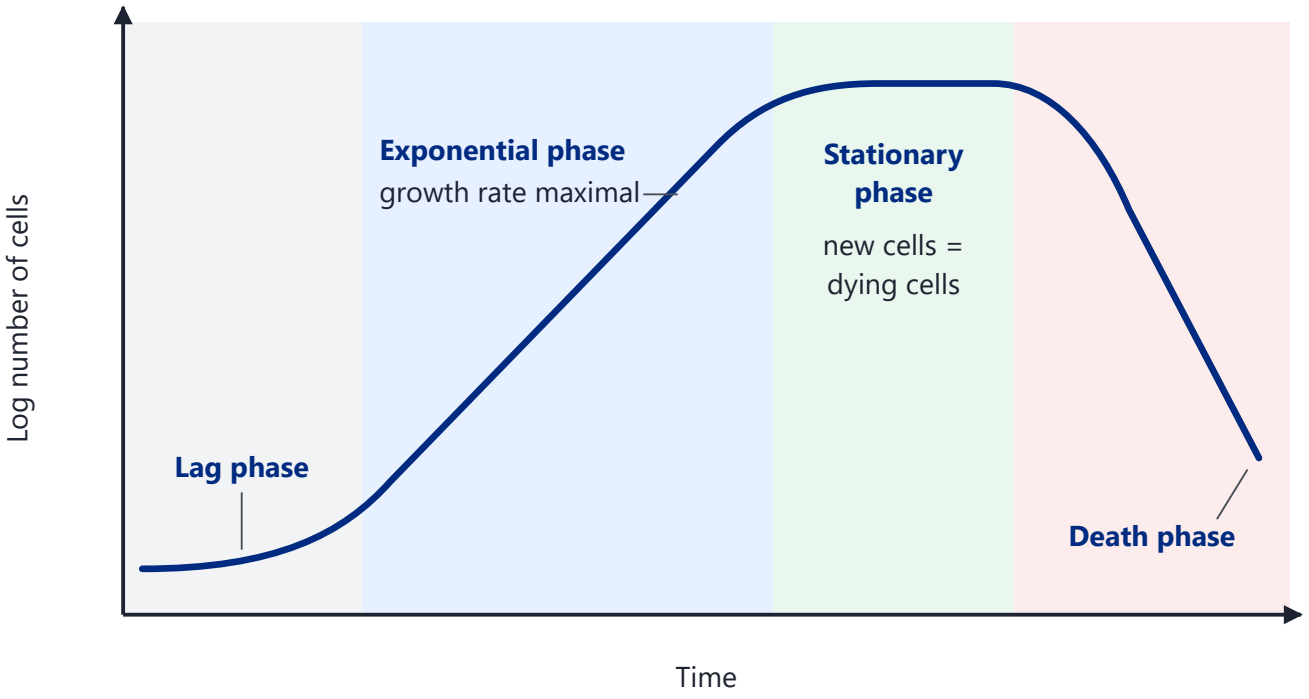
Dilution plating works the same way: divide the colony count by the volume spread on the plate and multiply by the dilution factor.

Method	Advantages	Disadvantages
Haemocytometer	Quick and cheap; shows cell shape; a stain such as trypan blue colours dead cells so they can be excluded	Very small cells are hard to see; moving cells must be immobilised
Dilution plating	Counts only living cells	Slow, needs incubation; a clump of cells gives only one colony; only cells that grow in the conditions provided are counted
Turbidity	Fast; repeated readings can be taken from one culture	Does not distinguish living cells from dead cells or other particles; needs calibration

4. Bacterial growth curves

Bacteria reproduce asexually by binary fission. The circular DNA is copied, the cell grows longer, and a cross-wall forms between the two copies, dividing the cell into two genetically identical daughter cells. The time between divisions is the generation time; in ideal conditions some species divide every 20 minutes, though most take longer.

When bacteria are put into fresh broth, the population passes through four phases. In the lag phase the cells are adapting, for example by making the enzymes they need, so numbers hardly rise. In the exponential (log) phase nutrients are plentiful and waste is low, so the cells divide as fast as they can and the population doubles every generation. In the stationary phase nutrients run short and toxic waste builds up, so the rate of division equals the rate of death and the number stays constant. In the death (decline) phase more cells die than are produced.



A bacterial growth curve: lag, exponential, stationary and death phases.

Cell numbers rise so fast that they are plotted on a logarithmic scale, on which the exponential phase is a straight line. During this phase the number of cells is given by $N_t = N_0 \times 2^{kt}$, where N_0 is the starting number, N_t the number after time t and k the growth rate constant, the number of generations per hour. Rearranging gives:

KEY IDEA

$k = (\log_{10} N_t - \log_{10} N_0) \div (\log_{10} 2 \times t)$. Use it only for the exponential phase.

Finding the growth rate constant

1. A culture grows from 1 cell to 1024 cells in 5 hours of exponential growth. List the values: $N_0 = 1$, $N_t = 1024$, $t = 5$ h.
2. $\log_{10} 1024 = 3.01$ and $\log_{10} 1 = 0$.
3. $k = (3.01 - 0) \div (0.301 \times 5) = 3.01 \div 1.505 = 2.0$ generations per hour.

Check: $1024 = 2^{10}$, so there were 10 generations in 5 hours, which is 2 per hour.

The graph below models a broth culture, and the triangle measures k from the gradient of the log graph.

CORE PRACTICAL

Core practical: growth in a liquid culture. Keep a yeast culture in glucose solution at constant temperature, stirred so cells stay suspended and nutrients and oxygen are spread evenly. At intervals, remove samples with a sterile pipette and measure absorbance in a colorimeter. Plot log absorbance against time, identify the lag and exponential phases, and calculate k using absorbance in place of cell number. Yeast is a safe, ethical choice because it has no nervous system.

5. Pathogens: how they enter and what stops them

Pathogens enter the body by several routes:

- **Inhalation:** droplets from coughs and sneezes carry pathogens; some droplets dry into tiny particles that stay in the air for a long time.
- **Ingestion:** in food or water contaminated with pathogens, often from faeces.
- **Inoculation:** directly through a break in the skin, such as a cut, an animal bite or a shared needle.
- **Vectors:** living organisms, such as insects, that carry a pathogen from one host to another.
- **Fomites:** objects such as towels, bedding and medical equipment that carry pathogens between people.
- **Direct contact:** touching, as in many skin infections and sexually transmitted infections.

The first line of defence is a set of barriers. The skin's outer layer is made of dead cells packed with the tough protein keratin, so pathogens cannot get through unbroken skin. Sebaceous glands secrete sebum, and sweat contains substances such as lactic acid, which make the skin surface acidic and inhibit microbial growth. The airways and other internal tubes are lined with cells that secrete mucus, which traps microorganisms, and cilia sweep the mucus away to be swallowed. Tears, saliva and mucus contain lysozyme, an enzyme that breaks down bacterial cell walls. Hydrochloric acid in the stomach kills most microorganisms in food.

The skin and gut are also covered by harmless bacteria, the skin flora and gut flora, which are adapted to live there. These commensal bacteria compete with pathogens for space and nutrients and outcompete them, and some secrete substances that inhibit pathogens.

6. Tuberculosis and HIV

Tuberculosis

Tuberculosis (TB) is caused by *Mycobacterium tuberculosis*, a rod-shaped aerobic bacterium with a thick, waxy coat that protects it from drying, disinfectants and digestion by macrophages. It spreads mainly in droplets from coughs and sneezes. In the lungs, macrophages engulf the bacteria, but many survive and multiply inside them. The body walls off the infected area in a granuloma (tubercle) of macrophages, other immune cells and dead tissue, and the bacteria can survive there for years as latent TB.

If the immune system is weakened, for example by HIV, the bacteria break out and active TB develops. Destruction of alveoli reduces the surface area for gas exchange, causing breathlessness, and damaged tissue and blood vessels cause a persistent cough with blood in the sputum, along with fever, tiredness and weight loss. TB also

suppresses the immune system by reducing T cell numbers. Death can follow from respiratory failure, spread to other organs, or opportunistic infections.

HIV and AIDS

HIV is a retrovirus. Its glycoprotein gp120 binds to CD4 receptors on helper T cells and the envelope fuses with the cell surface membrane, releasing the capsid into the cell. Reverse transcriptase copies the viral RNA into double-stranded DNA, which integrase inserts into the host's DNA as a provirus, where it can stay latent for years. When the T cell is activated, the host's RNA polymerase transcribes the provirus. The viral RNA acts as mRNA for viral proteins and as the genome of new viruses, which bud from the cell, taking some of its membrane as their envelope.

Infected helper T cells are destroyed as viruses bud out and by killer T cells. Helper T cells are needed to activate B cells and killer T cells, so as their number falls the specific immune response fails. A person can be HIV positive for years with few symptoms; AIDS is the late stage, when the helper T cell count is very low and opportunistic infections such as TB and pneumonia cause death. HIV is treated with a mixture of drugs, because it mutates rapidly and a strain resistant to one drug is unlikely to resist them all.

7. The non-specific response

If a pathogen gets past the barriers, non-specific responses act against any foreign material, without needing to recognise the particular pathogen.

- **Inflammation:** damaged tissue causes mast cells to release histamine. Histamine makes arterioles dilate, so more blood flows to the area, which becomes red and hot, and makes capillary walls more permeable, so fluid, phagocytes and antimicrobial proteins leave the blood. The tissue swells.
- **Phagocytosis:** phagocytes such as neutrophils and macrophages engulf a pathogen into a vacuole (phagosome). Lysosomes fuse with it and release enzymes that digest the pathogen. A macrophage then displays the pathogen's antigens on its surface and becomes an antigen-presenting cell.
- **Lysozyme:** an enzyme in tears, saliva and mucus that breaks down bacterial cell walls, so the bacteria burst.
- **Interferon:** a protein released by virus-infected cells. It makes neighbouring cells produce proteins that inhibit viral protein synthesis, so the virus cannot spread easily, and it helps to activate immune cells.

8. The specific immune response

An antigen is a molecule, usually a protein or glycoprotein on a cell surface, that the body recognises as foreign and that triggers an immune response. Lymphocytes carry receptors of one shape, so each recognises only one antigen. B cells mature in the bone marrow, and T cells mature in the thymus.

From infection to antibodies and killer T cells

1. A macrophage engulfs the pathogen and displays its antigens on its surface. It is now an antigen-presenting cell (APC).

2. A T helper cell with a complementary receptor binds to the antigen on the APC. It is activated and divides by mitosis into a clone of active T helper cells and T memory cells.

3. A B cell with a complementary receptor binds the same antigen, takes it in and presents it, becoming an APC itself. An active T helper cell binds to it and releases cytokines.

   The B cell cannot be fully activated without these cytokines, which is one reason the first response is slow.

4. The activated B cell divides to form a clone of B effector cells and B memory cells. B effector cells differentiate into plasma cells, which secrete antibodies; this is the humoral response.

   Plasma cells have extensive rough endoplasmic reticulum to make antibody protein and many mitochondria to supply ATP.

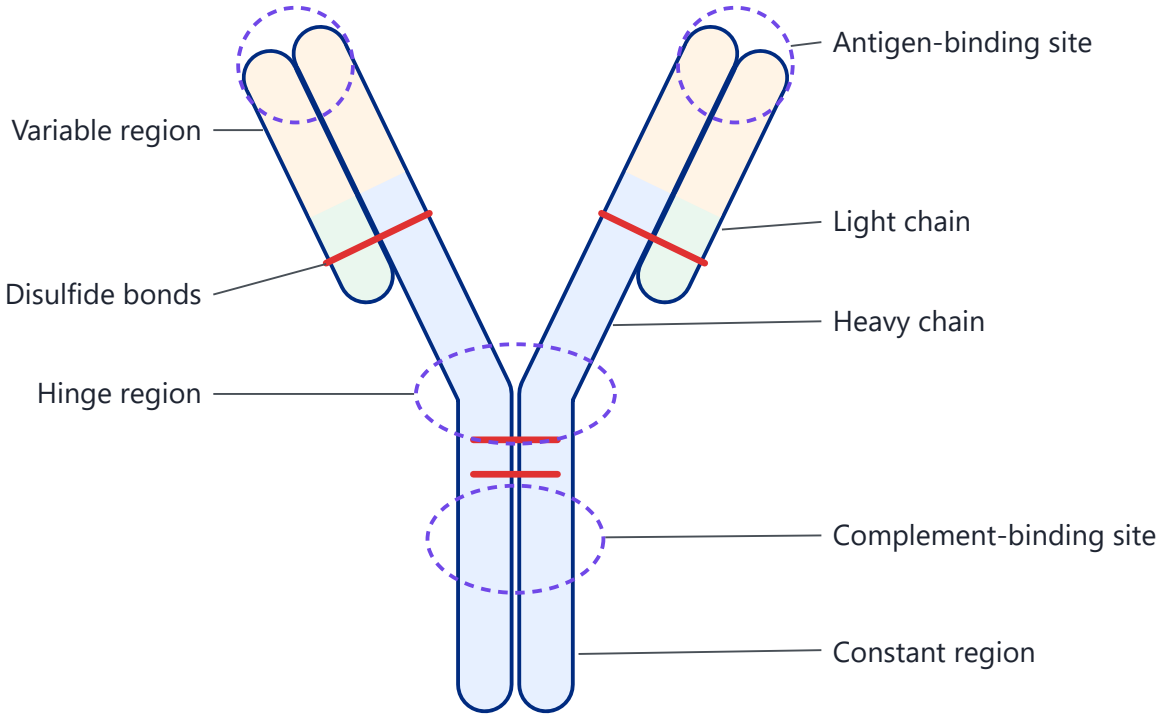
5. Cytokines from T helper cells also activate T killer cells with complementary receptors, which divide into active T killer cells and T memory cells.

6. In the cell-mediated response, T killer cells bind to body cells displaying the antigen, such as virus-infected cells, and release chemicals that make pores in their membranes, so the cells burst. Phagocytes engulf the released pathogens.

Cell	Role
B effector cell	Differentiates into plasma cells, which secrete antibodies
B memory cell	Persists for years; divides rapidly into plasma cells if the antigen returns
T helper cell	Releases cytokines that activate B cells and T killer cells and stimulate phagocytes
T killer cell	Destroys infected body cells displaying the antigen
T memory cell	Persists; gives a fast T cell response if the antigen returns

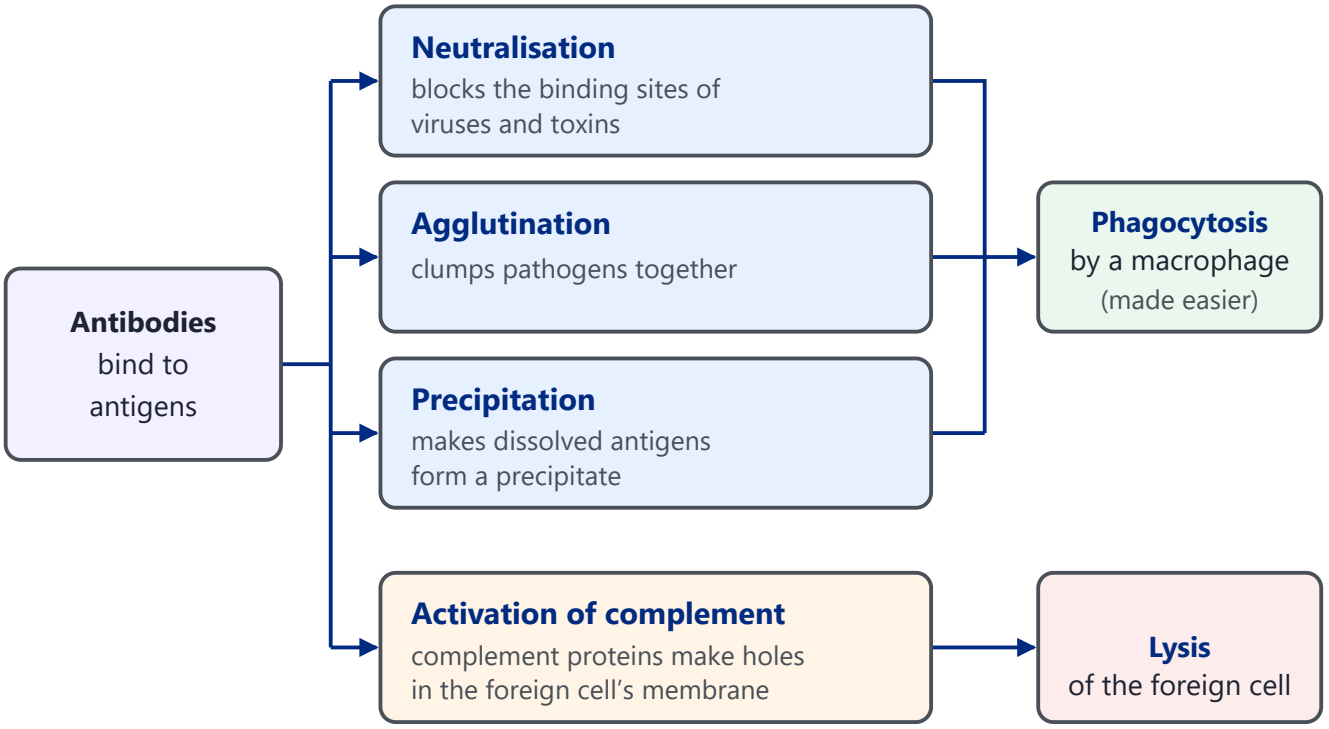
Antibodies

An antibody (immunoglobulin) is a protein with quaternary structure: two long heavy chains and two short light chains held together by disulfide bonds. The tips of the arms are the variable regions, whose shape differs between antibodies and is complementary to one antigen. The rest is the constant region, the same in antibodies of one class, which binds to receptors on phagocytes. A flexible hinge lets the two arms bind antigens at different distances apart.



An antibody: the variable regions form the antigen-binding sites; the constant region interacts with immune cells.

Antibodies neutralise viruses and toxins by blocking their binding sites, clump pathogens together (agglutination) and make dissolved antigens precipitate, all of which make phagocytosis easier. They also activate complement proteins, which make holes in a foreign cell's membrane so that it bursts.



Antibody binding leads either to phagocytosis or to lysis of the foreign cell.

Primary and secondary responses

On first exposure to an antigen, only a few lymphocytes have the matching receptor, and they must be activated, divide and differentiate before antibodies appear. This primary response takes about 10 to 17 days, so symptoms usually develop. Memory cells from the primary response persist for months or years. If the same antigen returns, they divide rapidly into plasma cells and T killer cells, so the secondary response is faster and produces far more antibody, and the pathogen is usually destroyed before symptoms appear. The person is immune.

The graph below compares the two responses, and the ways in which immunity can be acquired.

9. Natural, artificial, active and passive immunity

In active immunity the body makes its own antibodies and memory cells, so protection takes time to develop but lasts. In passive immunity ready-made antibodies are received; they protect at once, but no memory cells form and the antibodies are soon broken down.

Type	How it is acquired	Memory cells?
Natural active	Infection by the pathogen	Yes: long-lasting
Artificial active	Vaccination with a dead or weakened pathogen or its antigens, which triggers the immune response without the disease	Yes: long-lasting
Natural passive	Antibodies from the mother across the placenta and in breast milk, especially colostrum	No: short-term
Artificial passive	Injection of antibodies made in another person or animal	No: short-term

10. The evolutionary race between pathogens and hosts

Hosts and pathogens exert selection pressures on each other. Pathogens with mutations that help them avoid the immune response survive and reproduce, while hosts that resist infection better are more likely to survive and reproduce. Pathogens have short generation times and huge populations, so new mutations arise often and they evolve quickly. The many ways pathogens evade the immune system are evidence for this evolutionary race.

- **HIV:** destroys the helper T cells that coordinate the immune response; mutates rapidly, so its antigens keep changing and antibodies and memory cells no longer recognise them; and hides as a latent provirus inside host cells.
- ***M. tuberculosis*:** its waxy coat resists digestion, so it survives inside macrophages where antibodies cannot reach it; it can interfere with antigen presentation; and it can stay latent inside granulomas for years.
- **Antibiotic resistance:** the use of antibiotics is a further selection pressure that pathogens have evolved to resist.

11. Antibiotics

Antibiotics are chemicals, originally made by microorganisms, that kill or inhibit bacteria. Bactericidal antibiotics, such as penicillin, kill bacteria. Bacteriostatic antibiotics, such as erythromycin, stop them multiplying, giving the immune system time to destroy them; if the drug is removed, the bacteria can grow again. The effect also depends on dose: at high doses some bacteriostatic antibiotics kill many bacteria. Broad-spectrum antibiotics act against many kinds of bacteria, including useful gut bacteria; narrow-spectrum antibiotics act against a few.

Antibiotics target features that bacteria have but human cells lack or that differ. Some inhibit peptidoglycan synthesis, so the weakened wall bursts by osmotic lysis. Others bind to 70S ribosomes and stop protein synthesis without affecting human 80S ribosomes. Others damage the membrane, inhibit DNA replication or transcription, or block specific enzymes. Antibiotics do not work against viruses, which have no wall, ribosomes or metabolism of their own.

CORE PRACTICAL

Core practical: effect of different antibiotics on bacteria. Using aseptic technique, spread a bacterial culture over nutrient agar and place on it discs soaked in different antibiotics, plus a control disc soaked in sterile water. After incubation, measure the diameter of the clear zone of inhibition around each disc and calculate its area with πr^2. A larger clear zone indicates a more effective antibiotic at that concentration.

How resistance spreads

Resistance arises by random mutation, for example one that makes the wall impermeable to the drug or lets the bacterium make an enzyme that breaks it down. The antibiotic is a selection pressure: susceptible bacteria die, while resistant ones survive, multiply and pass on the resistance allele (vertical transmission), so its frequency rises. Resistance genes on plasmids can also pass between bacteria, even of different species, by conjugation through pili (horizontal transmission).

Hospital-acquired infections

Hospital-acquired (nosocomial) infections, such as MRSA (methicillin-resistant *Staphylococcus aureus*) and *Clostridium difficile*, are caught in hospital. Patients are often weak or have wounds or medical devices, many antibiotics are used, and pathogens pass easily between patients on hands, equipment and bedding. Understanding these causes has led to codes of practice.

- **Prescribing:** use antibiotics only when needed, never for viral infections or "just in case"; choose a narrow-spectrum drug suited to the infection; make sure patients finish the course; and rotate the antibiotics used.
- **Infection control:** hand washing, gloves and aprons; isolating infected patients; screening patients on admission; no jewellery or outdoor clothing on wards; safe disposal of dressings and needles; sterilising bedding and equipment.

EXAM TIP

Two reasons not to use a broad-spectrum antibiotic unnecessarily: it kills useful bacteria as well as pathogens, and it puts selection pressure on many species at once.

12. Quick checks

QUICK CHECK

In which phase of a bacterial growth curve can the growth rate constant k be calculated?

- Lag phase
- Stationary phase
- Exponential (log) phase
- Death phase

The formula assumes that the population doubles at a constant rate, which is only true in the exponential phase, where log cell number rises in a straight line. In the stationary phase division and death are balanced, so the line is flat.

QUICK CHECK

Which feature is found in HIV but not in tobacco mosaic virus?

- A lipid envelope and reverse transcriptase
- RNA as the genetic material
- A protein capsid
- 70S ribosomes

Both viruses have RNA and a protein capsid, but only HIV has a lipid envelope and reverse transcriptase. No virus has ribosomes.

QUICK CHECK

Which statement correctly describes latency in a virus such as lambda phage?

- The virus bursts the host cell at once.
- The viral DNA is destroyed by the host.
- The virus makes its own ribosomes.
- The viral DNA is inserted into the host DNA and copied with it, but its genes are not expressed.

In the lysogenic pathway the viral DNA becomes part of the host chromosome, is copied each time the cell divides and is kept inactive by a repressor protein. Bursting the cell at once describes the lytic pathway, the commonest wrong answer.

QUICK CHECK

Which cells secrete antibodies?

- T helper cells
- Plasma cells, which develop from B effector cells
- T killer cells
- Macrophages

Activated B cells divide into B effector and B memory cells, and B effector cells differentiate into plasma cells that secrete antibodies. T helper cells release cytokines, not antibodies, which is the commonest confusion.

QUICK CHECK

Why do antibodies a baby receives in breast milk give only short-term protection?

- The antibodies are digested before they reach the blood.
- The antibodies are the wrong shape for the baby's pathogens.
- This is passive immunity: no memory cells are made, and the antibodies are broken down.
- Breast milk contains antibiotics rather than antibodies.

In natural passive immunity the baby's own immune system is not activated, so no memory cells form, and the received antibodies are gradually broken down. Lasting protection needs active immunity.

QUICK CHECK

How does a bacteriostatic antibiotic differ from a bactericidal one?

- It stops bacteria multiplying rather than killing them.
- It works only against viruses.
- It kills every bacterium in the body.
- It acts only on human cells.

Bacteriostatic drugs prevent multiplication, often by blocking protein synthesis, so the immune system can remove the bacteria. Bactericidal drugs kill them. No antibiotic works against viruses.

A2 · Unit 4 · WBI14 Energy, Environment, Microbiology and Immunity

Decomposition and forensics

Forensic biology uses DNA to link people to evidence, and uses the predictable changes in a dead body to estimate how long ago death occurred. This page explains how the polymerase chain reaction copies DNA, how fragments are separated and profiled, and how body temperature, rigor mortis, decomposition and insects are combined to estimate the time of death.

1. Copying DNA with the polymerase chain reaction

Samples from a crime scene, such as a drop of blood or a few cells on a swab, often contain too little DNA to analyse. The polymerase chain reaction (PCR) makes millions of copies of chosen regions of DNA in a few hours. The DNA is mixed in a tube with plenty of all four DNA nucleotides, a pair of primers, a heat-stable DNA polymerase and a buffer, and placed in a thermal cycler that changes the temperature automatically. Each cycle has three steps.

One cycle of PCR

1. Denaturation: the mixture is heated to about 95 °C.

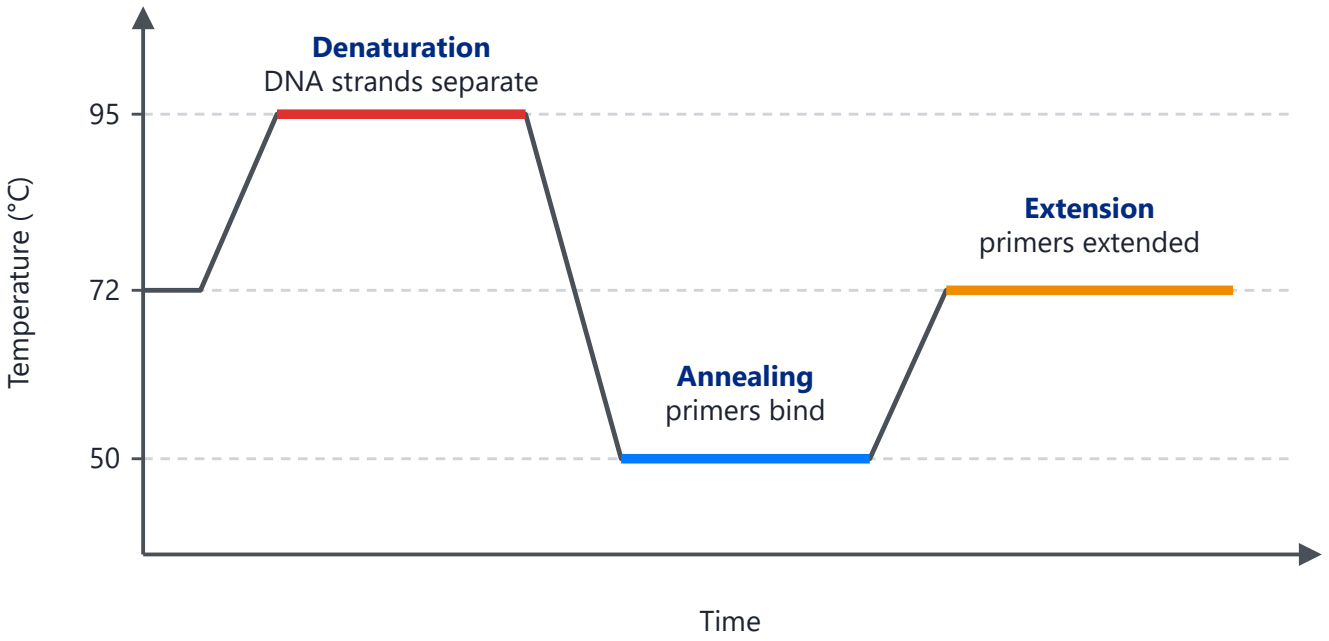
   This breaks the hydrogen bonds between complementary bases, so the double helix separates into two single strands that can act as templates.

2. Annealing: the mixture is cooled to about 50 to 60 °C, so the primers bind (anneal) by complementary base pairing to the start of the target region on each strand.

   A primer is a short single strand of DNA, about 20 bases long. DNA polymerase can only add nucleotides to an existing strand, so it needs a primer to start.

3. Extension: the temperature is raised to about 72 °C, the optimum of the polymerase. The enzyme adds free nucleotides to the end of each primer, making a new strand complementary to each template.

4. Each original molecule has now become two, so the number of copies doubles with every cycle. About 30 cycles are usually run.



The temperature profile of one PCR cycle: denature at about 95 °C, anneal at about 50 °C, extend at about 72 °C.

The polymerase comes from a bacterium that lives in hot springs, such as *Thermus aquaticus* (Taq polymerase). Because it is heat stable, it is not denatured in the 95 °C step, so it does not need to be replaced every cycle. Its high optimum temperature also means extension can happen at a temperature above the annealing temperature, which makes each cycle efficient. Human DNA polymerase works best at about 37 °C and would be denatured, because its active site would change shape, at PCR temperatures.

Working out the number of PCR cycles

1. Each cycle doubles the number of DNA molecules, so after n cycles one molecule has become 2^n molecules.
2. To make 128 molecules from one, set $2^n = 128$.
3. $128 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7$, so 7 cycles are needed.

   Check: after 6 cycles there would be only 64 molecules.

2. Separating DNA fragments by gel electrophoresis

Gel electrophoresis separates DNA fragments according to their length. Restriction enzymes cut the DNA at specific base sequences, producing fragments of different lengths. The fragments are loaded into wells at one end of a slab of agarose gel covered with a buffer solution, and an electric current is passed through it. Phosphate groups make DNA negatively charged, so the fragments move towards the positive electrode. The gel acts as a

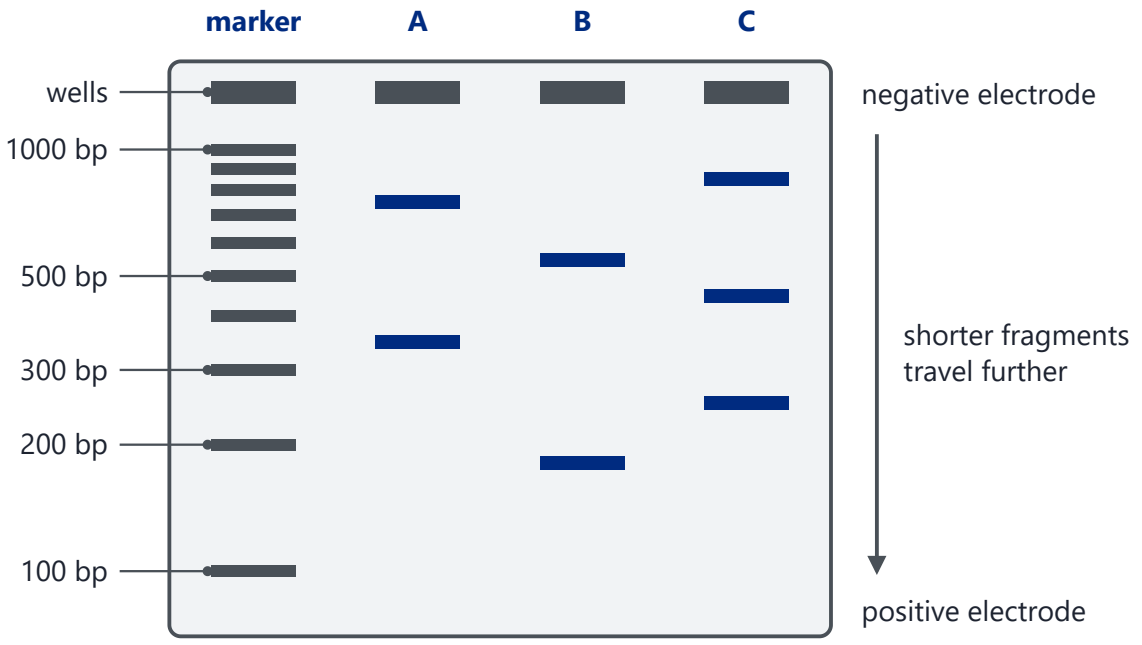
sieve: shorter fragments move through its pores more easily and travel further in a given time, so after the run the fragments are spread out by size, with the shortest furthest from the wells.

COMMON MISTAKE

Electrophoresis separates DNA fragments by size, not by charge. Every nucleotide carries the same negative charge, so the charge for each unit of length is the same for all fragments; it is the sieving effect of the gel that makes shorter fragments move faster.

The DNA bands cannot be seen directly. In the traditional method, the DNA is made single-stranded and transferred from the gel onto a nylon or nitrocellulose membrane placed on top of it (a Southern blot). The membrane is then treated with DNA probes: short, single-stranded pieces of DNA with a base sequence complementary to the regions of interest, labelled with a radioactive isotope or a fluorescent marker. The probes bind only where there is a complementary sequence. With radioactive probes, an X-ray film placed on the membrane darkens where the probes are; fluorescent probes glow under ultraviolet light.

A DNA ladder, or marker, is run in one lane alongside the samples. It contains fragments of known lengths, so the length of each sample band can be estimated by comparing how far it has travelled.



The marker lane gives fragment lengths, so the bands in samples A, B and C can be sized and compared.

3. DNA profiling

Most human DNA does not code for proteins. Some of these non-coding regions contain short base sequences repeated one after another, called variable number tandem repeats (VNTRs). The number of repeats at each

VNTR locus varies greatly between people. Each person inherits two copies of each locus, one from each parent, so each locus gives one or two bands. Because many loci are analysed, the chance of two unrelated people having the same pattern is extremely small; only identical twins share all their VNTR lengths. Repeats come in two sizes: microsatellites, or short tandem repeats, have units of 2 to 6 bases, and minisatellites have longer units of about 10 to 100 bases.

Making a DNA profile

1. Collect samples from the crime scene and from suspects, such as blood, saliva or hair roots, and extract and purify the DNA.
2. Amplify the DNA by PCR so there is enough to analyse.
3. Cut the DNA with restriction enzymes to produce fragments that contain the VNTRs.
4. Separate the fragments by gel electrophoresis, make them visible with labelled probes, and compare the band patterns of the samples.

If a suspect's bands match those from the crime scene at every locus, the samples very probably came from the same person; if any band does not match, the suspect is excluded. In a paternity test, every band in a child's profile must have come from either the mother or the father. Bands in the child that are not in the mother's profile must be present in the real father's profile, so a man who lacks them cannot be the father.

The more closely related two individuals are, the more bands their profiles share, so profiling can also show genetic relationships between organisms, both plants and animals. For identifying species, scientists compare the base sequence of a short standard region of DNA, a DNA barcode. In animals a region of a mitochondrial gene is usually used; in plants this region changes too slowly to tell species apart, so regions of chloroplast DNA are used instead.

- **Primer:** a short single strand of DNA complementary to the start of the region to be copied.
- **Restriction enzyme:** an enzyme that cuts DNA at a specific base sequence.
- **VNTR:** a non-coding region in which a short sequence is repeated a variable number of times.
- **Probe:** a labelled single strand of DNA that binds to a complementary sequence.

EXAM TIP

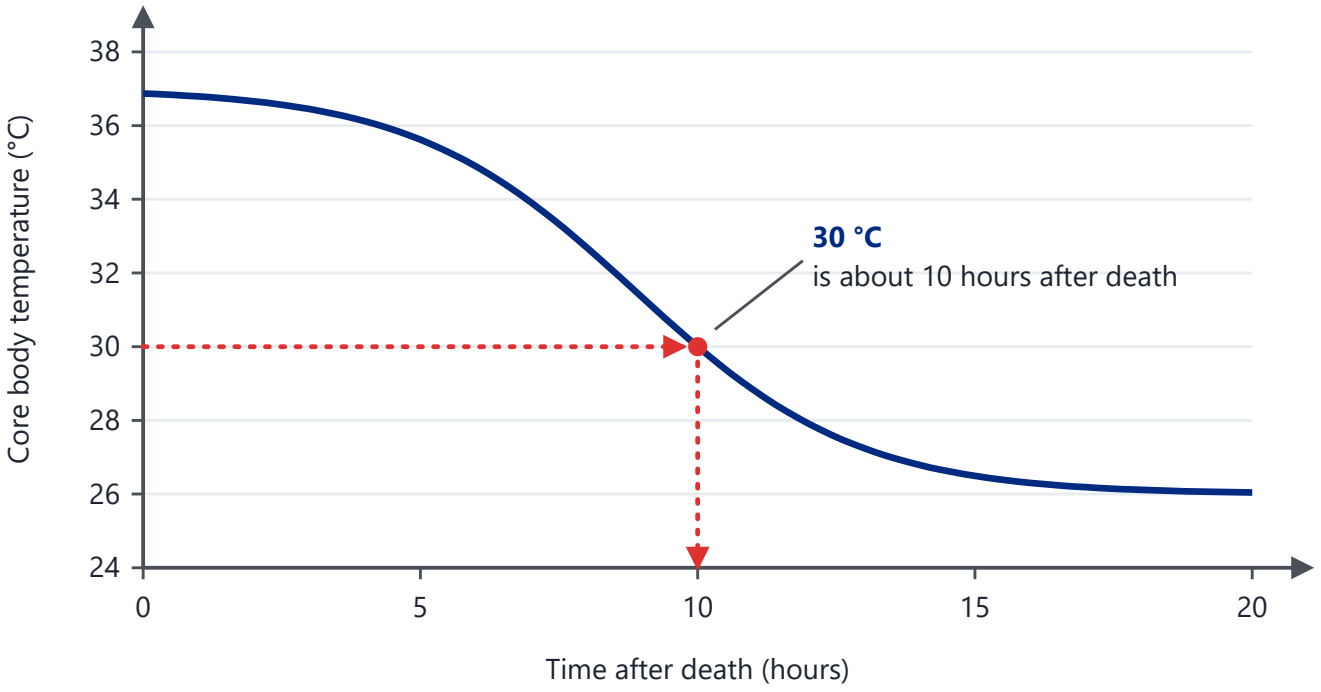
Describe a profile comparison in terms of bands in the same positions on the gel. Do not say that the people have the same genes: profiling compares the lengths of non-coding repeats at a few loci.

4. Estimating the time of death

After death, a mammal's body changes in an orderly sequence, and each change happens at a fairly predictable rate. Forensic scientists use body temperature, the degree of muscle contraction (rigor mortis), the extent of decomposition, the stage of insect development and the stage of succession of organisms on the body. All of these depend on temperature, because temperature controls the rate of enzyme-controlled reactions, so the ambient temperature must always be recorded. The estimates are combined with witness and other evidence to give the most likely time of death.

Body temperature

In life, the body is kept at about 37 °C by heat released in metabolic reactions such as respiration. After death these reactions slow and stop, and the body loses heat to its surroundings by radiation, conduction and evaporation of water. This cooling is called algor mortis. The core temperature, measured in the rectum or with a probe in the abdomen, stays almost steady for a short time, then falls steadily, and finally levels off at the temperature of the surroundings, giving a roughly sigmoid curve.



A cooling curve: reading across from a measured core temperature gives an estimate of the time since death.

Reading a cooling curve

1. Measure the core temperature, say 30 °C, and the ambient temperature as soon as possible, and choose a curve drawn for similar conditions.

   The rate of cooling depends on the difference between body and ambient temperature, and the ambient temperature may change over time.

2. Find 30 °C on the temperature axis and read across to the curve.

3. Read down to the time axis: about 10 hours after death for the curve shown.

Several factors change the rate of cooling. Clothing and body fat insulate the body and slow cooling. A body curled up has a smaller exposed surface area than one stretched out, so it cools more slowly. A large body has a smaller surface area to volume ratio than a small one, so it also cools more slowly. A body in water cools faster than one in air, because water conducts heat away better, and moving air cools a body faster than still air. A body found indoors usually cools more slowly than one outdoors.

KEY IDEA

If a clothed body is matched to a curve for a naked body, the estimated time since death will be too short, because the clothed body cooled more slowly. A body found in moving air would give an estimate that is too long if a still-air curve were used.

The cooling curve below is drawn for the conditions you choose, so you can read off an estimate and see how each factor changes it.

Rigor mortis

Rigor mortis is the stiffening of the muscles after death. When the heart stops, cells are starved of oxygen, so aerobic respiration stops. Anaerobic respiration continues for a while, producing lactic acid, which lowers the pH and inhibits enzymes, including those that make ATP. Muscle needs ATP to break the cross-bridges between myosin heads and actin. Without ATP the myosin stays bound to actin, so the muscle cannot relax and becomes fixed in position.

Rigor usually starts 2 to 4 hours after death in the small muscles of the face and neck and spreads down the body to the larger muscles, reaching full effect after about 6 to 8 hours. Small muscles stiffen before large ones, and not all muscles are fully rigid at the same time, so the pattern of stiffness in several muscles is used. Later, the muscles relax again as their proteins are broken down during decomposition. Rigor develops sooner if there was little ATP in the muscles at death, for example after hard exercise, and faster at higher temperatures, because the chemical reactions are faster. Physical damage, immersion in water, burning, electrocution, clothing and air movement all affect its progress.

Decomposition

After death, cells begin to digest themselves, a process called autolysis. Enzymes from the gut start to break down the gut wall and nearby cells, and lysosomes in dying cells burst and release digestive enzymes. Autolysis is faster in mild warmth and slowed by cold, and by intense heat that denatures the enzymes. Soon afterwards, bacteria from the gut and the airways multiply and spread through the tissues, digesting them; this is putrefaction. Microorganisms secrete enzymes that break down proteins, carbohydrates and fats, and their respiration releases carbon dioxide and other gases, returning carbon and other elements to the environment.

The visible signs follow a pattern, although the timings vary greatly with temperature:

- **About 36 hours:** green discolouration of the skin over the abdomen.
- **36 to 72 hours:** the discolouration spreads over the body and darkens to reddish green.
- **About 72 hours:** the discolouration becomes purple-black.
- **About a week:** gases from bacterial action, such as hydrogen sulfide, methane and carbon dioxide, make the body bloat.
- **More than a week:** the gas escapes and the body deflates and shrinks.

Decomposition is faster in warm conditions, because the enzymes of the body and of bacteria work faster. A body on the surface decomposes faster than a buried one, because insects, other decomposers and scavengers can reach it more easily. Body temperature and rigor are most useful in the first 48 hours; after that, the state of decomposition and insect evidence are used.

Forensic entomology

Forensic entomology is the use of insects to estimate the time of death. Blowflies arrive very soon after death and lay eggs in wounds and in openings such as the nose and mouth. The eggs hatch into larvae (maggots), which feed, grow and moult through several stages before pupating and emerging as adults. At a given temperature, each species takes a known time to complete each stage; development is faster in warm conditions and slower in cold. By identifying the species, measuring the length or finding the stage of the oldest larvae, and knowing the ambient temperature, scientists work out how long ago the eggs were laid, and so roughly when the person died.

COMMON MISTAKE

Insect evidence gives the time since the eggs were laid, not directly the time since death. It assumes that flies laid eggs soon after death and that the temperature was fairly constant. If the body was hidden or wrapped, egg laying may have been delayed.

The timeline below shows the stages of a typical blowfly at a constant temperature, with the stage reached drawn to scale.

Succession on a body

A dead body is a rapidly changing habitat. As it decomposes, its condition changes, and different species of insects and other organisms are attracted in turn; each group also changes the body in ways that attract the next. This

sequence of communities is a succession, and it follows a predictable order, so the community present when the body is found indicates how far decomposition has gone and so roughly how long ago death occurred. Insects can also show that a body has been moved: if it carries species that live only outdoors or in a particular habitat, it must have been there at some point after death.

The timeline below shows a typical sequence on a body left in the open; real timings vary from case to case.

Combining the evidence

The best estimate comes from several independent methods that agree. Core and ambient temperatures are useful for roughly the first day, rigor mortis for the first day or two, the signs of decomposition for later periods, and insect development and succession for days to weeks. Each method must be corrected for the temperature, clothing, position, size and surroundings of the body.

5. Quick checks

QUICK CHECK

What is the purpose of heating to 95 °C in each PCR cycle?

- To let the primers bind to the template
- To give the polymerase its optimum temperature
- To break hydrogen bonds so the DNA strands separate
- To join nucleotides to the primers

At 95 °C the hydrogen bonds between complementary bases break, giving single strands that act as templates. Primers bind at the lower annealing temperature, and the polymerase extends them at about 72 °C. The commonest wrong answer confuses denaturation with annealing.

QUICK CHECK

How many PCR cycles are needed to produce 128 molecules from one DNA molecule?

- 6
- 8
- 64
- 7 ✓

After n cycles there are 2^n molecules, and $2^7 = 128$. Six cycles give only 64 and eight give 256.

QUICK CHECK

Why do shorter DNA fragments travel further in gel electrophoresis?

- They pass through the pores of the gel more easily. ✓
- They carry less negative charge.
- They are attracted to the negative electrode.
- They are denser, so they sink further.

Charge per unit length is the same for all fragments, so separation depends on the sieving effect of the gel. DNA is negative and moves towards the positive electrode.

QUICK CHECK

Why do muscles become stiff in rigor mortis?

- Bacteria produce gas inside the muscles.
- ATP is no longer made, so myosin heads stay attached to actin.
- Calcium ions are removed from the muscle.
- The muscles fill with water as cells burst.

After death, aerobic respiration stops; anaerobic respiration produces lactic acid, the pH falls and ATP production stops. ATP is needed to detach myosin from actin, so the cross-bridges stay locked. Gas from bacteria causes bloating, which happens much later.

QUICK CHECK

A clothed body is found indoors. If a cooling curve for a naked body is used, how will the estimated time since death compare with the true time?

- It will be too long.
- It will be correct.
- It will be too short.
- It cannot be estimated at all.

Clothing insulates the body, so it cools more slowly than a naked body. Reaching the measured temperature took longer than the naked curve suggests, so the estimate is too short.

QUICK CHECK

Why must the ambient temperature be recorded when maggots are used to estimate the time of death?

- Maggots only feed on bodies above 37 °C.
- The temperature tells you which species of fly arrived first.
- Cold kills all fly eggs.
- The rate of insect development depends on temperature, because it is controlled by enzymes.

Each stage takes a known time only at a given temperature: warmth speeds development and cold slows it. Without the temperature, the age of the maggots cannot be converted into a time since the eggs were laid.